The NIST / ANSI seed-selection search

Comprehensive investigation report

Twenty-one curves. Six reconstructed streams.
Eleven qualifying omissions. One unresolved historical decision.

Central finding: the seed bytes are reproducible, but the original rule or process that selected the published curves has not been recovered. Many specific arithmetic and implementation explanations fail. Some exploratory fits and field-specific patterns remain observations, without the provenance needed to identify the historical mechanism.

Evidence population	Historical criterion	Current research boundary
10 published + 11 omitted	UNRESOLVED	No further predicate search

Prepared from the retained October 16-28 investigations and the November 6 Mattsson reconciliation. This report distinguishes arithmetic facts, documentary findings, configured-software behavior, engineering controls, exploratory fits, and unknown history.

Includes a source catalog, exact curve records, and embedded evidence attachments. Prepared for the research owner; this is a synthesis, not an independent replication of every archived computation.

DECISION RECORD

Executive findings

The search did not identify the criteria used by the original program. It did explain the published prime-field seed bytes and establish exact identities for the 21-curve comparison. The reconstructed omissions are qualifying stream positions, not documented rejection events.

Finding	Supported conclusion
Seed provenance	All ten published prime-field seeds match clock-initialized rand48 streams. Six initialization timestamps account for them.
Ordinary validation	The 21 curves pass the retained order, nonsingularity, non-anomalous and bounded MOV checks. The final pass verifies 63 original/opposite-b/twist order certificates.
Curve arithmetic	Twist strength, exact embedding degree, conductors, class-group tests, auxiliary-input divisors and fixed workflow proxies do not identify the historical split.
Specific leads	P-224 cubic j / 3-division splitting is a real local pattern. An exploratory auxiliary-prime-199 two-rule classifier fits all 21 labels. Neither has established original-program use.
Historical software	Period sources and later point-counting reconstructions expose real branches and possible faults, but no tested, authenticated acceptance path explains all eleven omissions.
ECPP boundary	A period proof-search exhaustion mechanism is reproducible on an induced control. Retained cohort-wrapper engineering records execute zero named orders; this is not a cohort result.
Mattsson reconciliation	All 19 overlapping curve identities reconcile. One P-224 sign change is isomorphic; SEA list/cutoff differences are convention and formula differences.
Completion status	Historical objective unresolved. Mattsson follow-up PR #3129 is open at the last verified check, with all 22 hosted checks successful.

Evidence: [R48] [R58] [R27] [R41] [R54]

NAVIGATION AND CLAIM LABELS

How to read this report

The main narrative follows the chain from recovered seed bytes to candidate arithmetic, contemporary workflows, source archaeology, and the closing checks. Appendix A catalogs every retained report used in this synthesis, including superseded intermediate records. Appendix B prints exact identities and arithmetic parameters for all 21 curves. Appendix C lists primary external sources and the final evidence boundary.

Label	Meaning in this report
Verified / derived arithmetic	An exact archived computation, certificate or independent arithmetic check supports the stated finite result. This PDF does not rerun the scientific producer.
Retrieved documentary fact	A specific source was obtained and read in the archived study. A dated citation is distinguished from a retrieved original draft.
Configured-software result	Behavior of a specified recovered or reconstructed program under its recorded build and inputs; original NIST use remains unproven.
Engineering control	A compatibility, framing, mock, or correctness observation. It does not count as a candidate-level scientific result.
Exploratory fit	A relationship found after labels or correlated features were available. Exact fit alone is not historical causation or an independently predictive rule.
Unknown / censored	Missing inputs, incomplete factorization, timeout, unavailable source, or unobserved execution. It is not recoded as acceptance or rejection.

Reference tags such as [R01] link to the exact report path in a pinned repository snapshot. Appendix A gives the path and archival synopsis; the embedded source manifest gives each file SHA-256. Chronology follows recorded report dates, while archive-capture dates, PDF metadata, reconstructed timestamps and publication dates retain different meanings.

Important correction: an older source-search note and one later refresh said the November 1997 draft had not been obtained. The recovered-draft report supersedes that statement. The 103-page November draft is available; the separately cited June 1996, January 1997 and June 1997 versions remain missing as full draft texts.

Evidence: [R09] [R47]

WHAT THE SEARCH ATTEMPTED

1. Scope and exact workload

The objective was to explain why eleven prime-field seed positions that pass the reconstructed eligibility screen were not published, while ten others were. The user raised possible transfer, j-invariant, endomorphism and CM/conductor explanations and suggested that two rules might operate together. The work therefore examined both curve properties and contemporary generation/validation behavior.

The fixed comparison has ten published curves and eleven reconstructed omissions, across six streams and five field sizes. The four fields with both labels contain nineteen curves: five at 192 bits, four at 224 bits, eight at 239 bits and two at 256 bits. P-384 and P-521 are published singletons. Results on nineteen rows cannot silently be promoted to a complete 21-row result.

The exact model is y^2 = x^3 - 3x + b over F_p. Each row retains p, b, n, the SHA-derived parameter c, full seed, timestamp, one-based ordinal and full curve UID. The trace is t=p+1-n. The symbol c is not the trace. In the canonical records the coefficient relation is c*b^2 = -27 mod p.

The eleven windows are not proven to have been visited by the original program. Extra RNG calls, restarts, a separate point generator, review choices, or a flawed counting or certification path could prevent a simple block-by-block history. Retrospective mathematical eligibility does not demonstrate an original rejection for weakness.

Fixed-population accounting

Population	Published	Omitted	Use
Complete cohort	10	11	All six streams
Same-field comparison	8	11	Four fields with both labels
Mattsson external table	8	11	Excludes published P-384 and P-521

Evidence: [R67] [R51] [R54]

SEED BYTES VERSUS SELECTION

2. Recovered seed generation

The public reconstruction identifies consecutive 20-byte seed blocks from the high byte of successive rand48 states, initialized by srand48(timestamp). In integer form, x_next = (0x5DEECE66D*x + 0xB) mod 2^48, with initial state (timestamp << 16) | 0x330E; each output byte is x_next >> 40. Equivalently, use lrand48() >> 23. This regenerates all ten prime-field published seeds.

Timestamp / UTC start	Published positions	Other qualifying positions
863711343 / 1997-05-15 15:49:03	239v1: 2185; v2: 2320; v3: 3048	36, 1562, 2491, 2682, 2863
863966734 / 1997-05-18 14:45:34	192v1: 139; v3: 5228; v2: 6013	2198, 5405
871002724 / 1997-08-08 01:12:04	P-521: 1136	None in the reported cohort
871696354 / 1997-08-16 01:52:34	P-256: 1842	876
872952608 / 1997-08-30 14:50:08	P-384: 2688	None in the reported cohort
891607927 / 1998-04-03 12:52:07	P-224: 7026	777, 1485, 5736

The seed-reconstruction result is much stronger than a remembered phrase story: it predicts the exact bytes. It does not identify the selection rule or the number of attempted timestamp restarts. The timestamps are mathematically reconstructed initialization values; they are not surviving wall-clock logs.

The other fairly clear extra restriction is bitlen(n) <= bitlen(p). It is not n < p: published prime239v2 has p < n < 2^239. The eleven omissions are what remain under the particular retrospective root/order-width screen, not all positions admitted by the original near-prime standards text.

The broader record also reconstructs four binary-field seed streams and six May-1997 base points with FreeLIP, while B-163 and the four later NIST prime-field base-point recipes remain unresolved in the cited disclosure. Those ancillary findings do not change the 21-row prime-field selection workload.

Seed provenance evidence: [R70]

Evidence: [R62] [R51]

NO POST-ENDPOINT ADDITIONS

3. The 21-row cohort

Field	Ordinal	Canonical label	Published?
239	36	skip-863711343-36	no
239	1562	skip-863711343-1562	no
239	2185	prime239v1	yes
239	2320	prime239v2	yes
239	2491	skip-863711343-2491	no
239	2682	skip-863711343-2682	no
239	2863	skip-863711343-2863	no
239	3048	prime239v3	yes
192	139	prime192v1	yes
192	2198	skip-863966734-2198	no
192	5228	prime192v3	yes
192	5405	skip-863966734-5405	no
192	6013	prime192v2	yes
521	1136	secp521r1	yes
256	876	skip-871696354-876	no
256	1842	prime256v1	yes
384	2688	secp384r1	yes
224	777	skip-891607927-777	no
224	1485	skip-891607927-1485	no
224	5736	skip-891607927-5736	no
224	7026	secp224r1	yes

P-192 / prime192v1, P-256 / prime256v1, secp224r1 / P-224, secp384r1 / P-384, and secp521r1 / P-521 are aliases in these records. The 192-bit ordinal 5228 is v3, while ordinal 6013 is v2. Appendix B provides full seeds and UIDs so names alone are never the identity key.

Evidence: [R67] [R48]

ANSI AND NIST NEED NOT BE IDENTICAL

4. Standards chronology and version effects

A complete 103-page ANSI X9.62 working draft with a 17 November 1997 cover was recovered. Its PDF was encoded in December 1997 and archived later; those metadata do not notarize every circulation date. It already includes the seven published 192-, 239- and 256-bit prime examples, including P-256. Its generation procedure supplies no additional documented gate distinguishing the eleven omissions.

Dated material	What was obtained	What remains unknown
21 June 1996 draft	Citation in NIST SP 800-15	Full draft and parameter tables
15 January 1997 draft	Citation in USPS criteria	Full draft and parameter tables
11 June 1997 draft	Citation and unnamed example in a 1998 implementation paper	Full draft; exact identity of that example
17 November 1997 draft	Full text, seven exact published seed examples, generation procedure	Original generator and decisions
20 September 1998-cover text	Full later standards text	Date and purpose of each editorial change

There is a concrete rounding difference: the 1997 seed formula uses ceil(log2 p); the 1998-cover version uses floor(log2 p). The frozen replay finds that the earlier formula matches all 21 archived coefficients, while the later printed formula matches only 8/21 (4 publications and 4 omissions). It changes the curves for thirteen seeds. This discrepancy must be handled as a version issue, not as evidence that those thirteen were historically rejected.

The 1997 near-prime procedure also allows smooth cofactors and an operator-supplied minimum order. Retained certified records yield at least 69 full-prime-order seed positions satisfying its numerical checks, and at least 99 positions after including 30 near-prime cases. The latter is a lower bound: 10,083 other positions have censored exact-order outputs. These populations differ from the chosen 21-row retrospective screen and are not counts of historical attempts.

Evidence: [R09] [R17] [R02] [R07]

SIMPLE ARITHMETIC EXPLANATIONS

5. Root choices, twists and order filters

For the seventeen cohort rows with p=3 mod 4, the recorded prime-order b is the usual exponentiation root of -27/c. That convention retains nine publications and eight omissions. A lower-valued-root choice and parity choice also overlap both labels. At P-224, where p=1 mod 4, opposite b signs give isomorphic models and cannot supply an intrinsic distinction.

Every cohort curve has j different from 0 and 1728. The rational-twist classification therefore gives the ordinary pair of forms: the original and its nontrivial quadratic twist, rather than exceptional quartic or sextic twist families. A separate Sage rederivation agrees with the retained twist-order arithmetic.

Largest-twist-factor preference fails by explicit counterexample. For P-224 ordinals 777, 1485, 5736 and 7026, the largest twist-prime factors have 83, 122, 106 and 118 bits. Published P-224 is not the maximum. In the 239-bit field, omitted ordinal 36 has a 138-bit largest factor, between published v1 at 86 bits and v2 at 147 bits. A uniform monotone cutoff cannot recover all the labels.

A field-fixed minimum order r_min that retains every publication must also retain seven of the eleven omissions. It therefore cannot explain the entire split. Fixed order-shape/sparsity preferences likewise have counterexamples. A correct base-point search on a prime-order, cofactor-one group can use every nonidentity point, so failure to choose a generator is not an intrinsic curve exclusion.

Only one of the 21 true quadratic twists has prime order: published secp384r1. Requiring prime twist order would also discard nine published controls. Twist arithmetic is thus useful for testing specific rules but does not establish the original selection decision.

Evidence: [R51] [R68] [R63] [R08] [R65] [R48]

ATTACK PRECONDITIONS VERSUS CURVE SELECTION

6. Transfers and implementation input surfaces

The search explicitly checked pairing-transfer preconditions. For the nineteen curves in the four mixed fields, the exact multiplicative order k=ord_n(p) was computed and separately checked. The smallest k in that population has 189 bits; the smallest among the omissions has 190 bits. None meets the predeclared bounds 100, 1,000, 10,000 or 1,000,000. These particular low-embedding-degree rules reject none of the eleven omissions.

The twist-subgroup screen includes 25 certified twist factors of at least 80 bits across the full cohort. None satisfies p^k=1 mod ell for k from 1 through 1,000. This excludes the tested bounded transfer route on those factors; it is not a complete curve-security theorem or a proof that every transfer over every extension is impossible.

Order-neighbor divisors were also examined in relation to auxiliary-input algorithms. Such algorithms require secret-power auxiliary group elements beyond an ordinary ECDLP instance. The necessary inputs must not be assumed present in ECDSA or ECDH merely because n-1 or n+1 has a convenient divisor. The fixed divisor screen did not supply a common omitted-only opportunity, and the cited algorithm is later than 1997.

The conditional invalid-curve and singular-companion surfaces are shared within each field when a scalar implementation omits curve-membership checks and uses the same a=-3 addition formulas. The configured b does not restrict those alternate equations. Singular companions with b=+2 and -2 similarly depend on the shared field and a coefficient. These implementation-input surfaces do not separate the named candidates.

These results address specified attack preconditions and formula behavior. They neither recover a hidden transfer criterion nor establish a full security verdict. The unknown acceptance history should not be converted into evidence for a cryptanalytic weakness.

Evidence: [R45] [R69] [R19] [R53] [R66]

THE CONDUCTOR-OF-THREE CORRECTION

7. j-invariants and exact conductors

For these a=-3 models, j=6912/(4-b^2) mod p. The P-224 publication has cubic j and four linear factors in the 3-division polynomial. The three P-224 omissions have noncubic j and factor degrees 1 and 3. This is a genuine field-specific pattern. Published P-256 has noncubic j and the same 1,3 pattern as its omission, so a universal requirement that j be a cube fails.

The later signed-Frobenius audit gives the P-224 observation a more exact interpretation: the publication has scalar Frobenius -I on 3-torsion, while the three omissions have nonscalar Jordan action. The publication's quadratic twist consequently has full rational 3-torsion. This does not prove a universal equivalence between cubic j and a torsion pattern, or show that an original point counter needed a cube root of j.

Field	Omitted f_pi	Published f_pi
192 bits	1, 1	1, 1, 1
239 bits	1, 1, 1, 5, 1	1, 5, 1
224 bits	1, 1, 1	3
256 bits	1	1

Here Delta_pi=t^2-4p=f_pi^2*D_K, where D_K is the fundamental CM-field discriminant. The exact P-256 pair both have f_pi=1. The published curve with f_pi=3 in the retained same-field census is P-224, not P-256. A factor 3 in P-256's fundamental discriminant expresses ramification; it is not a conductor of three. An alternate model or curve setting must be identified explicitly before transferring a conductor claim.

The actual endomorphism-order conductor and the Frobenius-order conductor are different objects. The former divides f_pi; f_pi=1 forces the maximal order, but a nontrivial f_pi alone does not determine the exact endomorphism ring. The shared f_pi=5 on an omitted 239-bit row and published v2 is a direct counterexample to a simple conductor exclusion.

Evidence: [R58] [R43] [R44] [R27]

LATER REQUIREMENTS TESTED WITHOUT FITTED BOUNDS

8. CM class numbers and dated advice

A plausible mathematical motive for avoiding special CM structure is a class-number lower bound. The documentary search located h(O_K)>=200 in Baier's 2002 account and ETSI's July 2005 document, and h(O_K)>10^7 in RFC 5639 (March 2010). These are later sources; no retrieved artifact establishes their adoption by NIST in 1997.

The stronger direct certificate test uses the first 24 split rational primes and checks a sufficient discriminant inequality. The resulting distinct ideal classes certify h(K)>=2^24=16,777,216. All eleven omissions and eight published same-field controls receive that lower bound. P-384 and P-521 remain UNKNOWN_D_K for this particular method because certified fundamental discriminants were not imported. Unknowns are not counted as failures.

A subsequent operational test asks for a single ideal-class element of order at least 200. Taking the first odd split prime, no chosen class returns to the identity in powers 1 through 199 for any of those nineteen fields. Independent Python and GP computations agree on all 3,781 reduced powers per arm. Thus all eleven omissions satisfy both the class-number and element-order tests at the dated bounds.

The earlier genus-theory screen finds overlapping class-group 2-ranks: omitted ranks include 2,3,4,5,7,9, while publications include 1,2,3,4,6. Every field in the nineteen-row comparison has even class number. Neither parity nor a global rank threshold recovers the labels.

The report stages must be read in order. The initial source audit correctly said that the 10^7 bound had not yet been tested; the later certificate report completes the omission test. The earlier comparison to a modern externally claimed complete small-class-number list remains conditional on that external completeness claim. The direct nineteen-field certificates avoid that dependency. A large Frobenius-order class number must not be substituted for a lower bound on the maximal order without the proper conductor relation.

Evidence: [R31] [R29] [R30] [R52]

WHAT THE CONFIGURED REPLAYS ACTUALLY SHOW

9. RNG consumption and validation libraries

The search considered variable draw counts, unaligned consumption, wrappers, buffering and reseeding as alternatives to a curve-arithmetic gate. A model in which every validation call consumes the seed generator is not automatic: the reconstructed May base points use FreeLIP's separate RNG, while the curve seeds use libc rand48. The caller would have to connect the streams in a demonstrable way.

The FreeLIP prime-order replay classifies all eleven omitted orders and ten published orders alike under the specified zprobprime and zprime calls. Earlier native build failures tested zero cohort orders and remain retained. The successful build is a modern compatibility configuration, not an original system image.

A routine FreeLIP primality call can reproduce the seven-output prefix in all six May base-point recipes. With portable 26-bit limbs, zprobprime(v,7) or zprime(v,8,2), using v=p or v=n, yields all 24 proposed call/curve matches; 24 wrong-prefix or 30-bit controls match none. Several calls therefore explain the same observed prefix. This does not identify a unique historical call or explain the seed omissions.

Recovered March/August 1997 NTL source has a different literal consumption pattern: its default ProbPrime uses one small random base followed by ten full-size bases. Under 26-bit limbs the retained analysis derives at least 91 recurrence advances for 192-bit inputs and 111 for 239-bit inputs. Those bounds do not match a literal seven-output prefix. Source similarity is not enough to replace an executable call trace.

The pre-1997 FreeLIP audit checks 21 orders, two limb sizes, four seeds and A/A repeats. All 168 unique configured primality calls pass; the tested coefficient inversions and square roots pass as well. Square-root calls use no FreeLIP RNG draws, twenty-round primality uses twenty calls, and the independently seeded rand48 stream remains unchanged. All 756 recorded inverse aliasing/version configurations pass.

No retained source-bound wrapper, buffering or reseeding mechanism supplies a verified all-eleven omission explanation. This is not a proof that all possible RNG-consumption paths are excluded. The missing part is the original caller, its initialization and stream sharing, and its stopping or error policy.

Evidence: [R50] [R49] [R56] [R59]

ARITHMETIC PROXIES ARE NOT OLD EXECUTION TIMES

10. Point-counting cost and root branches

The strict Elkies-only trace-information proxy reaches its threshold earlier for omissions in 24 of 25 same-field comparisons. The P-256 pair reverses the direction. Adding Atkin information yields 20 earlier, three later and two tied comparisons, including a P-256 tie. Neither fixed proxy selects the eleven omissions. These are exact finite arithmetic comparisons, not measured 1997 runtimes.

The fixed two-cost Pareto check pairs the point-counting proxy with an n-1 certification proxy. Among sixteen preceding-omission/publication comparisons, omissions dominate publications in five pairs under one certificate-cost metric and thirteen under another. A strict policy requiring the publication to be better on both costs is contradicted. Changing the direction does not repair every field.

The all-root SEA audit follows every rational root at levels 3,5,13 in the available LiDIA build. Its final 63 cohort cells complete, with separate exceptional-j controls and independently checked arithmetic. The first attempt's absent table path and three aborting special controls are retained as execution observations. No omission-specific failed branch appears at those levels.

The CCR root-pairing reconstruction examines 147 curve/degree cells at levels 3,5,7,11,13,17,19. The corrected conversion recovers all 112 rational cyclic kernels in the favorable cells. None of the tested cohort projection collisions occurs. The literal printed conversion error instead loses higher-degree kernels for publications as well as omissions and therefore does not reproduce the selection split.

The modular-coordinate guard audit similarly finds that the bounded tested predicates pass all 21 curves. A real printed-formula discrepancy fails every cohort curve at degree 13 when interpreted literally. A bug that also loses the controls cannot serve as a recovered explanation of the historical labels.

The later full-counter build and explicitly reconstructed sign-handling arms remain version-bound observations. A modified or later source tree is not authenticated as the 1997 generator. Failed controls, missing tables, partial cell output and successful final markers with internal errors are not silently turned into curve rejection decisions.

Evidence: [R15] [R03] [R16] [R61] [R20] [R55]

A FIT EXISTS; ITS HISTORY IS UNPROVEN

11. Two-rule searches and the prime-199 fit

The earlier bounded library contains 1,913 Boolean predicates from 574 numeric/source columns. It yields 495 distinct complemented signatures on the 21 exact identities. The independent implementations evaluate 122,265 unordered signature pairs and all sixteen Boolean truth tables per pair: 1,956,240 cases. This library has zero robust or possible fits. That statement excludes this finite library, not every conceivable pair of criteria.

The signed-Frobenius extension adds new features and checks 1,281 cells: 21 curves times 61 odd primes through 293. The arithmetic implementations agree. This extended exploration finds exact fits, including the following at auxiliary prime 199: accept if C199 <= 32 and D199 <= 2048. C199 is a positive-half eigenvalue candidate-list metric; D199 is the minimum predicted irreducible-factor degree for the 199-division polynomial.

At ell=199	Publications	Omissions
Split/Elkies branch	4: C <= 20	7: C = 60
Nonsplit/Atkin branch	6: list skipped, D <= 1980	4: list skipped, D = 2200 or 2475
Combined displayed rule	10 retained	0 retained

The caps 32/2048 are post-observation representatives, not recovered constants. Any candidate cap 20..59 and degree cap 1980..2199 gives the same cohort labels. The search and calibration were exploratory; the descriptive capacity counts are not p-values or probabilities of an original choice. No held-out historical prediction establishes the rule.

The subsequent fixed-policy checks weaken a simple resource-limit interpretation. Applying the caps uniformly through 199 discards every publication. At 199 with the splitting degree unknown it retains only six publications. The measured later full-counter replay stops below 199 for all omissions; even published P-224 stops at 197. A callback at 199 cannot explain those particular replay omissions. The recovered source allocates 100 candidate slots at 199, not 32; a cap on every intermediate polynomial also contradicts controls.

This is why the full search cannot be summarized as either “nothing ever fits” or “the two criteria were found.” The exact fit is retained as a mathematical observation. The historical program, mandatory invocation, resource policy and cap provenance are missing.

Evidence: [R28] [R27] [R21]

ACTUAL DOCUMENTS VERSUS INFERRED LINEAGE

12. Historical source archaeology

Recovered sources include the relevant 1995 Mueller dissertation, Maurer's 1994 implementation thesis, Lehmann's 1994 modular-polynomial thesis, and a LiDIA package advertised as 1.3 in a May 1997 archive capture. These sources narrow the plausible workflow and expose actual function interfaces, relation tables and branch logic. None identifies the NIST/Reiter caller or its per-candidate decisions.

Mueller's two special fast-counting routes - supersingularity and ordinary isogeny classes containing j=0 or 1728 - occur on none of the 21 cohort rows in the fixed test. They are source-described counting shortcuts, not documented rejection criteria. The historical modular-polynomial interface accepts j directly; a universal cube-root-of-j prerequisite is not established.

A complete printed ell=5 polynomial is collated against a later table: G5(X,J)=(X^2+10X+5)^3-JX. All coefficient rows agree, including reduction modulo 101. This authenticates a mathematical table identity, not the original binary files, backend representation or NIST use. Historical native-word binary codecs differ across libI, LIP and GMP; source compatibility depends on backend and ABI.

The May 1997 package contains general polynomial degree routines but not the named standalone ECO caller. A later ChangeLog records a September 2001 splitting-degree fix. The exact faulty predecessor, patch and introduction date were not recovered. A 2019 root-wrapper fix must not be backdated into a 1997 root routine that lacked the same filtering call.

Static inspection finds an unguarded degree-one path into an invalid factor-list index. This is a source-level edge, not a native crash reproduction or a cohort rejection result. A conditional randomized-degree error bound is tiny under independent uniform field draws and valid inputs, but those assumptions do not prove the behavior of the actual deterministic RNG/caller. All 21 historical application statuses remain undetermined.

The recovered correspondence attributes the original programming to Bob Reiter. Solinas's retrospective phrase-hash recollection was qualified and the old archives had not been located. The exact rand48 byte reconstruction supersedes that recollection as a complete account of the prime seeds. The Reiter-Solinas tau-adic patent is a characteristic-two Koblitz setting, not an acceptance procedure for this prime-field cohort.

Evidence: [R26] [R25] [R23] [R24] [R22] [R47]

KEEP ENGINEERING CONTROLS SEPARATE

13. Prime-order proofs and ECPP

All nineteen same-field group orders have retained complete n-1 and n+1 factorizations. For every row, the product of n-1 factors at or below one million is smaller than sqrt(n), so a Pocklington proof using only those small factors is unavailable for every publication and omission. Mixed certificate-cost directions prevent a universal simple preference. This tests a workflow proxy, not the original proof engine.

The November 1997 draft describes probable-prime generation and an optional GKA certificate route. It does not require that every candidate pass one particular ECPP implementation. The recovered ECPP V3.4.1 distribution supplies a concrete finite-search failure path: exhaustion can return a value that its command-line caller labels composite. The package existed by a contemporary 1996 citation, but no artifact proves that Reiter used it.

A static 32-bit compatibility build passes 345 arithmetic comparisons and ordinary correctness controls. An elliptic-curve-only proof for 101 succeeds. With an intentionally empty discriminant catalog and both N+/-1 routes disabled, the same program labels prime 101 composite and exits zero; a missing catalog can instead crash. These are induced source-control outcomes, not observations on an omitted curve order.

The guarded cohort producer/checker implementation went through retained failed mock suites: parsing, identity mapping, JSON serialization and framing problems were fixed through a final nine-method passing mock suite. The archive addendum binds the released artifacts and source custody. The named cohort orders executed in those retained engineering reports remain zero. A ready wrapper, strict admission checker or passed mock suite does not establish the scientific comparison.

The ECPP branch is therefore incomplete at the historical-selection question. A certificate-search mechanism is plausible and executable on controls, but no retained result shows that it rejects exactly the eleven omitted orders and accepts the ten publications under an authenticated original configuration. This report does not launch that cohort experiment or imply that its approvals exist.

Evidence: [R57] [R14] [R42] [R41] [R38] [R37]

A FINITE SEARCH IS NOT HISTORICAL RECONSTRUCTION

14. Search extent, failures and uncertainty

The generator-extension study scans six additional slices of 1,024 blocks after the final published positions, separately from a 65,536-state P-256 coefficient-screen shard. These populations overlap in possible representations and are not additive counts of unique curves. The orbit shard applies a coefficient-root screen, not point counting or a weak-curve test. It is not an exhaustive search of the full 2^46 strict candidate-start space.

Under the strict twenty-draw block model, stepping by twenty in a full-period 48-bit recurrence gives 2^46 candidate-start states. Allowing arbitrary shifts gives a different ceiling. Neither bound includes unrestricted changes to the field, seed-generation algorithm or undisclosed arithmetic route. These bounds do not reveal the number of historical runs or establish a hidden weakness.

The broader exploratory feature and root/splitting screens are documented in the source catalog. Many overlap algebraically: trace, discriminant, power residues, local splitting and class-group features are correlated. Repeated nulls do not become independent evidence merely because several reports display them. Conversely, searching many thresholds and Boolean combinations makes exact in-sample fits easier to obtain.

Failures remain observations: the early twist factorization timeout was later resolved; FreeLIP builds initially failed before cohort calls; the base-point prefix build hit a watchdog before a portable build succeeded; ECPP mock suites failed before final engineering acceptance; SEA observers encountered missing tables and exceptional-control aborts; CCR attempts retained partial cells before a corrected run completed. The final fixed pass also retains a control-only GP precedence error repaired before its single cohort invocation.

Independent arithmetic checks are distinguished from independent software lineage. Sage and GP can share PARI internals. A separate Python group-law verifier strengthens order certification, but a modern library still does not become the historical implementation. Archive bytes, compiler flags, data tables, exact input identity, call framing, and caller behavior are separate provenance requirements.

Evidence: [R46] [R64] [R35] [R50] [R49] [R48]

USER-DIRECTED STOPPING BOUNDARY

15. The final fixed 21-curve pass

The closing instruction was to run the twist/b-root and MOV/anomalous/order checks once against all 21 curves without retuning, then stop further class-group, isogeny and predicate-library rounds. The retained run has one fresh PARI cohort invocation, zero retries, and a separate Python verifier. It certifies the original curve, opposite-b model and true nonsquare twist for each row: 63 curve-order certificates and 63 point witnesses.

Fixed property	Published true	Omitted true
Field/order primality, nonsingularity, seed relation	10/10	11/11
Original order certified; non-anomalous; order bit cap	10/10	11/11
MOV passes through 20 and through 100	10/10	11/11
Lower-valued b root	5/10	6/11
Even b root	6/10	6/11
p=3 mod 4 exponent root (where applicable)	9/9	8/8
Opposite-b order prime	2/10	3/11
Opposite-b order equals original order	1/10	3/11
True quadratic-twist order prime	1/10	0/11
Largest twist prime passes MOV 20 and 100	10/10	11/11

The exponent-root row is not applicable to four P-224 rows. The opposite-b equality row likewise reflects those four isomorphic models. No fixed flag or its complement is a perfect separator, and no combinations were searched in this closing pass. Imported complete twist factorizations were rigorously checked; no new factorization or SEA count was needed.

The order witness method verifies exact point order and that the Hasse interval contains a unique multiple of that order. The Python verifier checks every retained witness, factor-wise minimality and the Hasse uniqueness condition. The MOV no-hit result proves absence only in the stated range; the older nineteen-row exact embedding-degree result is a separate stronger calculation.

Closing interpretation: these checks do not identify the historical rule. Finite null tests do not prove that no possible arithmetic criterion exists. The work stops short of claiming that the selection process was purely non-arithmetic.

Evidence: [R48] [R45]

INDEPENDENT DATA RECONCILIATION, NOT ORIGINAL SOURCE

16. Mattsson comparison and publication

Mattsson's table states it was generated by Claude on 18 October 2026. The November 6 refresh downloads its commit-pinned and moving-main copies; both are 37,225 bytes with SHA-256 9fd4cf9a99482e44e7207b59aa831c58cec15682582a2932b65c04b3c6bc915c. An October 24 comparison already existed; the refresh replays it and binds all rows to the canonical 21-row record.

All nineteen seeds, ordinals, fields, a coefficients, c parameters, orders and signed traces match. Eighteen b coefficients match exactly. Published P-224 uses the opposite isomorphic sign; the retained explicit u witness verifies u^4=1 and b*u^6=-b mod p, giving (x,y) -> (u^2*x,u^3*y). The table covers all eleven omissions and eight publications, omitting published P-384 and P-521.

Every below-100 prime list agrees after recognizing that the external “Elkies” category includes ramified primes. There are 31 such entries across sixteen curves. Recomputing the retained strict signatures checks 3,173 Legendre symbols. The displayed literal threshold reproduces fifteen cutoffs, while the previously documented integer convention reproduces nineteen.

Curve	Reported	Literal formula
Omitted 239 / 36	179	181
Omitted 239 / 2682	157	179
Omitted 192 / 2198	149	163
prime239v2	191	197

The external formula is M>2^(L/2+2), while the exact output reconstruction uses M>2^(((L-1)//2)+2), including ramified primes. Our strict archive uses only +1 primes and M^2>16p. The generating code was not found in Mattsson's repository, so the reconstructed integer convention is not attributed as verified author code.

Follow-up PR #3129 publishes eight comparison artifacts at commit 134317a08335f64746eef21d507fd2219f213c45. All 22 hosted checks pass at the last verified observation; the PR remains open and unmerged then. Conductor claim T-5773 was released after recording that inherited Go checks belonged to its configured project rather than this Python artifact. Neither publication nor green CI resolves the historical criterion.

Evidence: [R54] [R11]

FINAL ASSESSMENT

17. What is established, and what is missing

Established: exact regeneration of the ten published prime-field seed bytes; exact identity and order evidence for the fixed cohort; numerous scoped arithmetic counterexamples; actual historical standards and library-source findings; bounded configured-software behavior; a preserved P-224 lead and an exploratory all-label fit; and a reconciled public nineteen-curve table.

Not established: that the original program visited all eleven reconstructed omissions; that it rejected them for weakness; that the prime-199 fit or a cube-root condition was used; that any particular recovered ECPP, FreeLIP, NTL or later LiDIA build was the NIST generator; or that the investigation has recovered the historically used criteria.

The main missing artifact is the original acceptance path: candidate visitation, validation calls, RNG ownership, proof/counting routines, resource limits, error handling and any human review. A run log or independently documented rule tied to exact candidates could be equally decisive. An earlier draft may constrain chronology, but a parameter table alone does not explain a rejection.

An arithmetic classifier would need a fixed, independently motivated rule, all-eleven/all-ten verification without label-dependent changes, and preferably a held-out historical prediction. Even that would require provenance before being called the actual 1997-98 mechanism. The user has closed further predicate searching; this PDF performs no new scientific search.

Corrections carried into this synthesis

The November 1997 draft is recovered, superseding older missing-draft notes. The conductor-three case in the exact same-field census is published P-224; both P-256 Frobenius conductors are one. The class-number source audit's initially unperformed stronger test is superseded by the nineteen certified field results, with two published unknowns retained. The Mattsson reconciliation is a replay of an existing comparison plus a fresh canonical identity binding, not the first analysis of that table.

Historical criterion: UNRESOLVED. The overall goal was marked blocked after repeated checks found no new original acceptance evidence. The completed comparison, publication and CI work are narrower finished deliverables; they do not substitute for the original objective.

The following appendices preserve the breadth of the search without treating every earlier report's prospective next step or administrative status as current. The source manifest and source-report bundle are embedded in this PDF for inspection offline. The exact curve JSON is also embedded.

Evidence: [R48] [R09] [R44] [R29] [R54]

ENTRIES 1-5 OF 70

Appendix A. Evidence catalog

Archival opening synopses below preserve each report's original scope. Earlier availability, planned-work and status statements may be superseded by the main narrative. These are internal research records, not authenticated original generator logs.

R01 | A pre-seed ANSI X9.62 draft citation

[R01] research/nist_seed_1996_preseed_draft_20261023/report.md

Date: 2026-10-23 UTC. Scope: Primary-source chronology for the missing prime-curve acceptance rule. Result: NIST Special Publication 800-15 cites an ANSI X9.62 ECDSA working draft dated 21 June 1996, before the reconstructed May-August 1997 seed streams. The cited draft's text and parameter tables were not obtained. This is evidence of a dated draft reference, not of what the original curve generator did.

R02 | The eleven are a narrow subset of the draft-compatible seed positions

[R02] research/nist_seed_1997_acceptance_20261024/report.md

Date: 2026-10-24 UTC. Question: If the recovered lrand48 streams are considered under the contemporaneous November 1997 X9.62 curve-selection procedure, does its listed mathematics isolate the eleven later named omissions? Answer: No. Of the already certified prime-order seed positions in the six prime-field stream prefixes, 69 pass the draft's listed numerical checks: 10 published and 59 unpublished. The eleven are the subset left by an additional retrospective root/bit-width screen. The 1997 draft does not state that ...

R03 | Atkin information changes the point-counting comparison without identifying a selector

[R03] research/nist_seed_1997_atkin_20261024/report.md

Date: 2026-10-24 UTC. Question: Did the earlier Elkies-only analysis omit enough contemporaneous point-counting information to hide a common reason for passing over the eleven named curves? Verified result: The frozen residue-density measure including Atkin information gives an earlier cutoff to an omitted curve in 20 of 25 same-field comparisons, a later cutoff in three, and a tie in two. The prior strict Elkies-only count was 24 earlier and one later. Including Atkin information makes the ...

R04 | Early X9.62 drafts are the next historical sources to recover

[R04] research/nist_seed_1997_draft_archive_20261023/report.md

Date: 2026-10-23 UTC. Scope: Read-only archival search for a contemporaneous X9.62 parameter record or acceptance rule. Result: Two primary IETF documents identify an ANSI X9.62 ECDSA working draft dated 17 November 1997. A NIST-hosted security policy also cites an X9.62 draft dated 31 August 1996. Neither draft's contents were obtained, so this search does not establish which curves either listed or why any candidate was passed over.

R05 | A January 1997 ANSI X9.62 draft citation before the recovered seed streams

[R05] research/nist_seed_1997_january_draft_20261023/report.md

Date: 2026-10-23 UTC. Question: Is there a dated X9.62 working-draft version between the June 1996 draft citation and the first reconstructed NIST/ANSI prime-curve seed streams in May 1997? Result: A USPS source cites an X9.62 ECDSA working draft dated 15 January 1997. The source does not include that draft's text or name any of the eleven omitted seed curves.

ENTRIES 6-10 OF 70

Appendix A. Evidence catalog

Archival opening synopses below preserve each report's original scope. Earlier availability, planned-work and status statements may be superseded by the main narrative. These are internal research records, not authenticated original generator logs.

R06 | A June 1997 X9.62 draft is cited beside a 192-bit example curve

[R06] research/nist_seed_1997_june_draft_20261023/report.md

Date: 2026-10-23 UTC. Scope: Primary-source search for the curve-selection record preceding the published NIST/ANSI prime-field parameters. Result: A 1998 implementation paper identifies an ANSI X9.62 draft dated 11 June 1997, earlier than the previously traced 17 November 1997 draft. The paper says its implementation used example curves from the ANSI draft, including a 192-bit prime-field example with a=p−3. It prints no seed, coefficient b, subgroup order, generator, candidate ordinal, or rejection reason for that example. ...

R07 | The 1997 draft admits more than the certified full-prime-order set

[R07] research/nist_seed_1997_nearprime_20261024/report.md

Date: 2026-10-24 UTC. Question: How many seed-derived curves in the retained positive exact-order records satisfy the November 1997 X9.62 draft's near-primality procedure with its example bound l_max=255? Verified result: 99 distinct seed positions: the prior 69 with prime full order and 30 additional positions with a 255-smooth cofactor and a proven large prime subgroup. All 99 have independent subgroup-point/Hasse certificates. This is a lower bound on the reconstructed prefixes' draft-compatible set, because 10,083 other seeded ...

R08 | A fixed minimum-order threshold cannot explain the eleven omissions

[R08] research/nist_seed_1997_order_threshold_20261024/report.md

Date: 2026-10-24 UTC. Question: Could the operator-supplied minimum subgroup order r_min in the 17 November 1997 ANSI X9.62 draft account for the eleven otherwise qualifying unpublished prime-field seed positions? Verified, conditional result: Under a field-fixed threshold that admits every published control in its field, seven of the eleven omitted prime-order curves must also pass. No such threshold separates all four fields. This is a post-observation, source-motivated falsification of one simple rule, not a discovery of ...

R09 | The recovered 17 November 1997 X9.62 draft

[R09] research/nist_seed_1997_original_draft_20261023/report.md

Date: 2026-10-23 UTC. Question: Does the original November 1997 working draft contain a rule that rejects the eleven otherwise qualifying seed candidates? Result: We recovered the full 103-page draft. Its stated prime-field curve-generation procedure gives no additional candidate predicate separating those eleven from the published curves. It already prints the seven published 192-, 239-, and 256-bit prime curves, including P-256. The actual generator's decision trace is still missing.

R10 | Did the published base points consume the seed stream directly?

[R10] research/nist_seed_1997_selection_extension_20261023/basepoint_stream/report.md

2026-10-24 UTC. The ten published prime-field base-point x coordinates do not occur as contiguous big-endian bytes of the reconstructed lrand48 high-byte stream in a bounded search covering each seed's entire prefix and 100,000 further bytes. This is a narrow implementation check, not a reconstruction of the original point generator or a curve-selection rule.

ENTRIES 11-15 OF 70

Appendix A. Evidence catalog

Archival opening synopses below preserve each report's original scope. Earlier availability, planned-work and status statements may be superseded by the main narrative. These are internal research records, not authenticated original generator logs.

R11 | Reconciling a public nineteen-curve table with strict Elkies arithmetic

[R11] research/nist_seed_1997_selection_extension_20261023/external_crosscheck/report.md

Date: 2026-10-24 UTC. Question: Does a newly published table of the same omitted and standardized curves independently reveal an all-eleven selection property, or correct the prior exact small-prime screen? Answer: It does neither. Its underlying seeds, group orders and traces match the archived curves, but its “Elkies” lists include ramified primes and its displayed cutoff formula does not exactly reproduce four of its own entries. One simple integer-threshold convention does reproduce all nineteen entries. The ...

R12 | NIST's 2014 seed-count statement and the remaining process gap

[R12] research/nist_seed_1997_selection_extension_20261023/nist_2014_process/report.md

Date: 2026-10-24 UTC. Question: Did NIST later state how many candidate seeds were tried when its prime-field curves were generated, or identify an extra historical acceptance criterion? Result: In its 2014 Committee of Visitors review, NIST could not state the number of attempted seeds. Its own presentation described a general repeat-until-secure-against-known-attacks process, with no per-candidate log or new mathematical test. This supplies a specific provenance limit; it neither explains the eleven omitted qualifying blocks nor ...

R13 | Does a sparse group order explain the passed-over curves?

[R13] research/nist_seed_1997_selection_extension_20261023/order_sparsity/report.md

2026-10-24 UTC. Result: No fixed preference for any of four predeclared group-order sparsity measures explains all eleven omitted prime-field seed classes under the assumption that the original program reached them. Several earlier omissions have sparser orders than the later published curve in the same field. This is a bounded test of a possible implementation preference, not a test of cryptographic weakness.

R14 | A possible 1997 certificate bottleneck, and its limits

[R14] research/nist_seed_1997_selection_extension_20261023/report.md

Date: 2026-10-23 local / 2026-10-24 UTC. Question: If the original program visited the eleven omitted prime-order seed curves, did a documented deterministic prime-order proof requirement reject them? Outcome: The 17 November 1997 draft includes an optional Goldwasser-Kilian-Atkin (GKA) certificate path. Its actual curve-generation path invokes probable-prime testing, without requiring a certificate. The 1998 text still permits alternative deterministic tests. Consequently the documented rule does not reject any of the eleven; a specific historical implementation might ...

R15 | A small-prime point-counting signature across the omitted curves

[R15] research/nist_seed_1997_selection_extension_20261023/sea_signature/report.md

Date: 2026-10-24 UTC. Question: Under the conditional premise that the original program reached all eleven omitted prime-order curves, could an unusually hard Schoof-Elkies-Atkin (SEA) point count have caused them to be skipped? Result: The frozen optimistic Elkies-only proxy points mostly in the opposite direction. Omitted curves reach its trace-information threshold at a smaller prime in 24 of 25 same-field comparisons. P-256 block 876 is the exception, so there is no universal cutoff separating all eleven. ...

ENTRIES 16-20 OF 70

Appendix A. Evidence catalog

Archival opening synopses below preserve each report's original scope. Earlier availability, planned-work and status statements may be superseded by the main narrative. These are internal research records, not authenticated original generator logs.

R16 | Can a two-cost 1997 workflow explain the skipped curves?

[R16] research/nist_seed_1997_selection_extension_20261023/workflow_pareto/report.md

Date: 2026-10-24 UTC. Question: If the original program evaluated all eleven omitted prime-order candidates, could it have published only candidates that were better on both an optimistic point-counting proxy and an n−1 primality-certificate proxy? Result: No for the fixed pair of proxies. Among the 16 same-field comparisons where the omitted candidate precedes a published curve, an omission dominates a publication in 5 pairs using required prime-factor bit length and 13 using the number of prime-power ...

R17 | A dated rounding change in the X9.62 seed formula

[R17] research/nist_seed_1997_selection_extension_20261023/x962_rounding/report.md

2026-10-24 UTC. The apparent contradiction between the later X9.62 prime-field seed formula and several published example curves has a precise version explanation. The 17 November 1997 draft, §F.3.3.b, PDF page 65, defines t = ceil(log□ p). The text with a 20 September 1998 cover, Annex A.3.3.2, PDF page 41, instead defines t = floor(log□ p). Both then clear the top bit of the first SHA-1 block. Each text repeats its respective rounding choice in the ...

R18 | A 1997 workflow reconstruction for the NIST/ANSI prime curves

[R18] research/nist_seed_1997_workflow_20261016/report.md

Date: 2026-10-16 Question: Which contemporary decisions plausibly produced the published curves, and which could explain eleven mathematically qualifying seed blocks passed over in the recovered streams? Status: Historical reconstruction and bounded comparison. The original acceptance code and logs have not been recovered.

R19 | Order-neighbor divisors do not mark the eleven skipped curves

[R19] research/nist_seed_aux_input_divisors_20261023/report.md

Date: 2026-10-23 UTC. Scope: The eleven omitted prime-field seed classes and eight published same-field controls, each identified by its exact EC1 UID in analysis.json. Verdict within this screen: No shared auxiliary-input divisor opportunity separates the labels. This result does not test ordinary ECDLP hardness or reconstruct a 1997 selection rule.

R20 | CCR root matching on all 21 NIST/ANSI curve candidates

[R20] research/nist_seed_ccr_root_matching_20261027/report.md

2026-10-27. T-5518. Observation report; no formal hypothesis decision. The missing historical selector remains unidentified. The corrected CCR root-pairing reconstruction recovers every one of 112 rational cyclic kernels across all 147 fixed-cohort cells. All eleven reconstructed omissions and all ten publications enter a favorable case at degree seven or higher. None exhibits the projected-coordinate collisions tested here.

ENTRIES 21-25 OF 70

Appendix A. Evidence catalog

Archival opening synopses below preserve each report's original scope. Earlier availability, planned-work and status statements may be superseded by the main narrative. These are internal research records, not authenticated original generator logs.

R21 | Exact selector capacity and period source recovery

[R21] research/nist_seed_ccr_root_matching_20261027/signed-frobenius-20261027/calibration-01/report.md

2026-10-27. Observation report for NIST-CLASSIFIER-CAPACITY-01, Conductor T-5518. This presents recorded data and a documentary comparison; no formal goal or hypothesis status is promoted. The actual historical selector remains unresolved. The ell-199 rule from the parent study retains all 10 publications and excludes all 11 reconstructed omissions across all six streams. Its narrow family is more selective than the broad two-feature search on the fixed reference populations below. These fractions are descriptive capacity, not p-values or ...

R22 | May 1997 degree routines: exact source audit and application limits

[R22] research/nist_seed_ccr_root_matching_20261027/signed-frobenius-20261027/period-eigenvalue-20261027/eco-code-recovery-01/degree-routines-01/report.md

28 October 2026 UTC, T-5518. Source-only observations and symbolic derivation; zero new native curve/control trials. The preceding class-group turn was progress. The full historical-selector goal remains active and unresolved. The source comparison authenticates a dated splitting-degree bug report and identifies an unguarded linear-input path. It does not identify the original caller or explain any of the eleven reconstructed omissions. All 21 full identities and six streams are retained in cohort-crosswalk.csv; historical application remains UNDETERMINED for ...

R23 | Recovered 1994 modular-polynomial source and May 1997 LiDIA bundle

[R23] research/nist_seed_ccr_root_matching_20261027/signed-frobenius-20261027/period-eigenvalue-20261027/eco-code-recovery-01/report.md

27 October 2026 PDT / 28 October UTC. T-5518, GOAL-NISTSEED-ab8949. The historical selector remains unresolved. This source-recovery round obtains Lehmann's actual thesis and an advertised LiDIA 1.3 bundle captured in May 1997. It checks the documented interface against the fixed eleven reconstructed omissions and ten publications across six streams using already completed evidence. There are zero new scientific trials and no original NIST/ANSI caller execution.

R24 | Original table witness and historical binary I/O

[R24] research/nist_seed_ccr_root_matching_20261027/signed-frobenius-20261027/period-eigenvalue-20261027/eco-code-recovery-01/table-artifacts-01/report.md

28 October 2026 UTC. T-5518 / GOAL-NISTSEED-ab8949. Historical selector unresolved. This source-only continuation verifies one printed 1994 coefficient table, recovers the binary interfaces in the May 1997 package, and checks a previously located history snapshot. It does not execute the original point counter or define a new curve-selection criterion. All eleven reconstructed omissions and ten publications remain in scope; there are zero new scientific trials.

R25 | The 1994 implementation thesis: actual limits and unresolved caller

[R25] research/nist_seed_ccr_root_matching_20261027/signed-frobenius-20261027/period-eigenvalue-20261027/implementation-sources-01/report.md

2026-10-27. T-5518, implementation-sources-01. Source review and a crosswalk to existing evidence; zero new scientific trials. The historical selector remains unresolved. This report changes no formal goal or hypothesis status. Retrieved: Markus Maurer's implementation thesis, with a title-page date of 29 July 1994, has been recovered from an archived university FTP mirror. This resolves the missing-Ma94 document identified in the preceding Muller1995 review. The thesis describes the actual eco program, source-file/function names, relation files, resource controls ...

ENTRIES 26-30 OF 70

Appendix A. Evidence catalog

Archival opening synopses below preserve each report's original scope. Earlier availability, planned-work and status statements may be superseded by the main narrative. These are internal research records, not authenticated original generator logs.

R26 | Muller1995 source routes on the complete NIST/ANSI cohort

[R26] research/nist_seed_ccr_root_matching_20261027/signed-frobenius-20261027/period-eigenvalue-20261027/report.md

2026-10-27. Observation report and primary-source review, NIST-MUELLER1995-ROUTES-01, T-5518. The historical selector remains unresolved. No formal goal/hypothesis status is promoted. New source progress: the original 1995 dissertation is now retrieved and read in the relevant sections. The previous round located its catalog but could not read the actual PDF. The public university link downloads through the in-app browser; the failed CLI/FTP retrievals are preserved separately. This updates availability, not the original selector's implementation provenance. New exact ...

R27 | Signed Frobenius: an exact two-rule separator and its source limits

[R27] research/nist_seed_ccr_root_matching_20261027/signed-frobenius-20261027/report.md

2026-10-27. Observation report; historical selector remains unresolved. This round adds exact evidence and changes no formal goal/hypothesis status. Observed: a two-predicate separator at auxiliary prime 199 retains all 10 publications and excludes all 11 reconstructed omissions across all six streams. It uses a source-derived eigenvalue candidate-list size and the minimum irreducible-factor degree predicted for the 199-division polynomial. The caps32/2048 are posthoc representatives of a range, not recovered historical constants. Let C199 be the maximum positive-half ...

R28 | Two-predicate audit of the fixed NIST/ANSI cohort - 2026-10-27

[R28] research/nist_seed_ccr_root_matching_20261027/two-rule-audit-20261027/report.md

These are exact observations and scoped comparisons. The historical selector remains unresolved; this report changes no formal hypothesis or goal status. Measured/derived result: 1,913 frozen Boolean predicates from 574 numeric/source columns give495 distinct complemented observation signatures on21 exact curve identities. Both implementations evaluated122,265 unordered signature pairs and all 16 truth tables per pair (1,956,240 distinct pair/function cases). There are zero robust fits and zero possible fits, including the explicit single-signature audit. Every pair has a ...

R29 | Dated CM class-number bounds: complete omission test

[R29] research/nist_seed_class_number_dated_bound_20261028/certificate-report-01/report.md

28 October 2026 UTC, T-5518. Observation report; no official research-state transition. Two frozen protocol versions and their original runs are retained.

R30 | A single class-group witness also passes on all eleven omissions

[R30] research/nist_seed_class_number_dated_bound_20261028/element-order-01/report-01/report.md

28 October 2026 UTC, T-5518; NIST-CLASS-ELEMENT-ORDER-01, run element-01. Observation report only; no official hypothesis or goal transition.

ENTRIES 31-35 OF 70

Appendix A. Evidence catalog

Archival opening synopses below preserve each report's original scope. Earlier availability, planned-work and status statements may be superseded by the main narrative. These are internal research records, not authenticated original generator logs.

R31 | Dated class-number advice: source audit

[R31] research/nist_seed_class_number_dated_bound_20261028/report.md

28 October2026 UTC, T-5518. No new scientific trials. The dated200 bound is already settled conditionally by the retained full21 comparison: ten publications and eleven omissions pass. The stronger10^7 bound has NOT been evaluated. This source audit must not be presented as the completed stronger test. Baier2002 section2.3.5 reports the German requirement h(O_K)>=200. ETSI TS102176-1 V1.2.1 July2005 clause6.1.2.3 and AnnexD.3 require the same200 lower bound on the maximal CM order. RFC5639 March2010 section2.1 item3 requires h(O_K)>10^7. ...

R32 | A low CM-field class number does not mark the eleven omissions

[R32] research/nist_seed_commonality_20261023/class_number_screen/report.md

Date: 2026-10-23 UTC. Status: Complete exact list comparison, conditional on the completeness claim of a newly released mathematical preprint. Question: If the original 1997-98 program evaluated all eleven qualifying but unpublished seed classes, could it have rejected them because their imaginary quadratic complex-multiplication (CM) fields had unusually small class numbers?

R33 | Could the eleven omitted curves share one CM field or a named isogeny relation?

[R33] research/nist_seed_commonality_20261023/cm_pair_screen/report.md

Date: 2026-10-23 UTC. Status: Exact structural result, independently checked; historical cause and cryptographic weakness remain unknown. Question: Under the stipulated premise that the original program evaluated all eleven otherwise qualifying but unpublished seed classes and rejected them for an unknown weakness, could that weakness be explained by their sharing one imaginary quadratic complex-multiplication (CM) field, or by a direct isogeny or twist relationship among the named curves?

R34 | A larger Frobenius splitting screen of the eleven omitted seed classes

[R34] research/nist_seed_commonality_20261023/isogeny_screen/report.md

Date: 2026-10-23 Status: Exploratory, independently checked arithmetic; no historical selection rule or ECDLP advantage identified. Question: Under the conditional premise that the 1997-98 program evaluated all eleven otherwise qualifying but unpublished prime-field seed classes, do they share a simple small-degree rational-isogeny property absent from the ten published classes?

R35 | The skipped-curve commonality search: small conductors and completed twists

[R35] research/nist_seed_commonality_20261023/report.md

Date: 2026-10-23 UTC. Status: Bounded, retrospective structural search. Question: If the original 1997-98 program evaluated all eleven reported qualifying but unpublished seed blocks, do they share a simple arithmetic trait absent from the ten published prime-field curves?

ENTRIES 36-40 OF 70

Appendix A. Evidence catalog

Archival opening synopses below preserve each report's original scope. Earlier availability, planned-work and status statements may be superseded by the main narrative. These are internal research records, not authenticated original generator logs.

R36 | Direct Coordinator access and period implementation follow-up

[R36] research/nist_seed_ecpp_compat_20261024/followup-20261024/report.md

2026-10-24. This record concerns runtime access, a bounded archival query and inspection of existing curve metadata. It records no named-order execution, scientific approval, common omitted-curve weakness or historical attribution.

R37 | Zero-run implementation archive completed

[R37] research/nist_seed_ecpp_compat_20261024/implementation-followup-20261024/archive-addendum/report.md

2026-10-25 UTC. Internal retrieved engineering custody record for GOAL-NISTSEED-ab8949, TASK-20261024-4f1bf4 and TASK-20261024-b17301. The verified Coordinator accepted completion at the declared static/mock implementation scope. The final V9 suite has nine methods, zero failures/errors and zero original-call attempts. The producer release binds all six declared artifacts. The isolated pushed snapshot is 68904e9b80c21785f1b17e6dfdb0e01b7aaf5a04, with parent 2786a4b7da61427725c66805d2ff7cb1f377e6b0 and fetched base ab0609ebee3d17527a51667591bda56f7cdece31. Its changed-path set is exactly the six implementation files and the precommit receipt. CORR-20261024-e1dbc8 resolves the self-reference: actual ...

R38 | ECPP cohort implementation: final guarded V9 result

[R38] research/nist_seed_ecpp_compat_20261024/implementation-followup-20261024/final-v9/report.md

2026-10-25 UTC. Engineering observations for TASK-20261024-4f1bf4 and prospective EXP-NISTSEED-6886da under GOAL-NISTSEED-ab8949. Named cohort orders executed: 0.

R39 | Cohort implementation status: three retained failed mock suites

[R39] research/nist_seed_ecpp_compat_20261024/implementation-followup-20261024/report.md

2026-10-24 UTC. Engineering observations only. Named orders executed: 0.

R40 | Full verifier-integration attempts through guarded wrapper V6

[R40] research/nist_seed_ecpp_compat_20261024/implementation-followup-20261024/v6-addendum/report.md

2026-10-25 UTC. No named order or legacy program executed. | Guarded capture | Syntax return code | Test methods | Reported errors | Original-call attempts | |---|---:|---:|---:|---:| | V4 | 0 | 7 | 23 | 0 | | V6 | 0 | 9 | 29 | 0 | Errors in these suites cascade from early implementation gates; they are not independent curve observations or independent defects. V4 stopped at the published P-224 exact-coefficient mapping. ...

ENTRIES 41-45 OF 70

Appendix A. Evidence catalog

Archival opening synopses below preserve each report's original scope. Earlier availability, planned-work and status statements may be superseded by the main narrative. These are internal research records, not authenticated original generator logs.

R41 | A 32-bit ECPP build reproduces the certificate-search failure label on an induced control

[R41] research/nist_seed_ecpp_compat_20261024/report.md

24 October 2026 UTC. Compatibility engineering and correctness controls; named historical orders executed: zero. The recovered ECPP 3.4.1 source now builds as a static 32-bit Linux executable. Its arithmetic passes 345 exact comparisons, its four ordinary small classifications are correct, and an elliptic-curve-only proof for 101 succeeds. With an intentionally empty discriminant catalog and both N±1 routes disabled, the same executable calls the prime 101 composite and exits zero. A missing catalog instead produces a ...

R42 | Recovered period ECPP code gives a concrete certificate-search failure hypothesis

[R42] research/nist_seed_ecpp_source_20261024/report.md

24 October 2026 UTC. Retrospective source audit; no named curve order executed. A preserved ECPP V3.4.1 package contains the original C implementation, arithmetic dependencies, discriminant catalog, factoring parameters and certificate checker. Two mirrors provide identical bytes. A thesis defended in October 1996 references this exact distribution filename. The source has a conditional path from finite certificate-search exhaustion to a zero return which its command-line caller labels “composite.” This is an identifiable software mechanism to test; ...

R43 | Exact Frobenius conductor on the omitted and published P-256 pair

[R43] research/nist_seed_exact_conductor_20261023/report.md

Date: 2026-10-23 UTC. Status: Exact first-stage arithmetic audit, not a historical selection result or full curve-security verdict.

R44 | Exact conductor census of the eleven omitted prime-field curves

[R44] research/nist_seed_exact_conductor_census_20261023/report.md

Date: 2026-10-23 UTC. Status: Complete exact arithmetic for the four fields containing both published and omitted seed-level classes. This is a retrospective structural comparison, not a recovered historical selection rule or a curve-security verdict.

R45 | Exact embedding degrees of the skipped NIST/ANSI seed candidates

[R45] research/nist_seed_exact_embedding_degree_20261023/report.md

Date: 2026-10-23 UTC. Scope: Eleven omitted prime-field seed candidates and eight published same-field controls. Status: Exact arithmetic result, separately checked in Sage. This is one attack-precondition screen, not a full curve-security verdict or a historical program replay.

ENTRIES 46-50 OF 70

Appendix A. Evidence catalog

Archival opening synopses below preserve each report's original scope. Earlier availability, planned-work and status statements may be superseded by the main narrative. These are internal research records, not authenticated original generator logs.

R46 | First extension and exact-state shard of the NIST prime-curve seed generator

[R46] research/nist_seed_exhaustive_20261016/report.md

Date: 2026-10-16. Status: Bounded search and exact curve checks. This is a census of the stated slices, not of the full generator orbit or the original 1997-98 decision process.

R47 | Direct-provenance refresh after the fixed arithmetic pass

[R47] research/nist_seed_final_checks_20261028/provenance-refresh-01/report.md

Observed 28 October 2026. This is a source-availability check, with zero new curve-arithmetic trials. The closing arithmetic report and its frozen results are unchanged. The original selection rule remains unresolved.

R48 | Final fixed arithmetic checks and requested stopping point

[R48] research/nist_seed_final_checks_20261028/report.md

User-directed closing pass, 28 October2026. Exactly one fresh PARI invocation covers all21 exact curves:10 publications,11 reconstructed omissions,six streams. No retuning or further predicate/class-group/isogeny rounds. All63 curve-order certificates pass: original, opposite-b and nontrivial quadratic twist for every curve. Separate Python arithmetic verifies 63 point witnesses, exact factor-wise point orders and the unique possible multiple in the Hasse interval. This freshly certifies the supplied original and twist orders without a new SEA count. PARI rigorously rechecks the ...

R49 | A routine FreeLIP primality call can supply the seven-output base-point prefix

[R49] research/nist_seed_freelip_prefix_20261024/report.md

Date: 2026-10-24 UTC. Question: Can a normal call in a period library account for the seven discarded random outputs in the reconstructed May 1997 base points? Verified configured-software result: Yes. In the tested portable 26-bit FreeLIP 1.0 build, zprobprime(v,7) and zprime(v,8,2), with either v=p or v=n, produce exactly the seven RNG advances and then all six published May Gx values. All 24 proposed primality-call/curve combinations match, as do the six manual-discard-seven baselines. The 24 wrong-prefix ...

R50 | FreeLIP 1.0 order-test replay on the skipped seed curves

[R50] research/nist_seed_freelip_primality_20261023/report.md

23 October 2026. This report records one narrowly specified software test. Its observation is that the tested FreeLIP 1.0 calls classify all 11 reconstructed omitted curve orders and all 10 published curve orders alike. It does not establish which program selected the historical curves or why. The user-stipulated premise that the original program reached and rejected all 11 remains unverified.

ENTRIES 51-55 OF 70

Appendix A. Evidence catalog

Archival opening synopses below preserve each report's original scope. Earlier availability, planned-work and status statements may be superseded by the main narrative. These are internal research records, not authenticated original generator logs.

R51 | What remains open in the NIST prime-curve seed-generation process?

[R51] research/nist_seed_generation_process_20261016/report.md

Date: 2026-10-16 Scope: The ten published prime-field curves and eleven other qualifying 20-byte blocks reported in six reconstructed 1997-98 srand48 streams. Status: Derived process constraints; the historical acceptance rule is unresolved.

R52 | CM genus ranks do not select the eleven skipped curves

[R52] research/nist_seed_genus_rank_20261023/report.md

Date: 2026-10-23 UTC. Scope: A retrospective exact-arithmetic comparison of the eleven qualifying but unpublished seed-level prime curves against eight published curves in the same four prime fields. This tests a possible class-group trait under the user's working premise that all eleven were evaluated and rejected; that premise is not historical evidence.

R53 | Could an invalid-curve weakness distinguish the eleven omissions?

[R53] research/nist_seed_invalid_curve_surface_20261023/report.md

Date: 2026-10-23 UTC. Status: Exact check complete for eleven omitted candidates and eight published controls over the same four field primes. This is a conditional analysis of unvalidated point input. The historical program's rejection criterion remains unknown.

R54 | Mattsson 19-curve identity and SEA-profile reconciliation

[R54] research/nist_seed_mattsson_comparison_20261106/report.md

Compared 6 November 2026. This is a fixed source comparison, not a selection-predicate search. An October 24 cross-check already existed in the worktree; this pass replays it against freshly retrieved source bytes and binds its rows to the later canonical 21-row identity record.

R55 | Canonical/Atkin coordinate guards and the CCR printed-formula correction

[R55] research/nist_seed_modular_coordinate_guards_20261027/report.md

2026-10-27. T-5518. Observation report; no formal hypothesis decision. The historical selector remains unknown. All 11 reconstructed omissions and all ten publications pass the bounded modular-coordinate predicates tested here, across all six streams. A newly recovered draft contains a real printed-formula discrepancy, but using those coefficients literally fails every cohort curve at degree 13 and therefore does not preserve the publications.

ENTRIES 56-60 OF 70

Appendix A. Evidence catalog

Archival opening synopses below preserve each report's original scope. Earlier availability, planned-work and status statements may be superseded by the main narrative. These are internal research records, not authenticated original generator logs.

R56 | Period NTL source narrows the seven-output prefix explanation

[R56] research/nist_seed_ntl_1997_source_20261024/report.md

24 October 2026 UTC. Retrospective source audit, without an NTL executable run. Four Internet Archive payloads match their CDX SHA-1 digests. The March and August 1997 source snapshots contain identical normalized bodies for ProbPrime(const ZZ&,long) and SqrRootMod. The default NTL prime test uses one small random base followed by ten full-size random bases. With 26-bit limbs, this requires at least 91 RNG recurrence advances for the 192-bit inputs and 111 for the 239-bit inputs. Those ...

R57 | Did easier prime-order certification select the published curves?

[R57] research/nist_seed_order_certificate_20261023/report.md

Date: 2026-10-23 UTC. Status: Complete exact factorization of n−1 and n+1 for all nineteen same-field published and omitted curves. This tests one possible 1997 workflow proxy; it does not reproduce the original program.

R58 | P-224 cubic j and 3-division splitting observation

[R58] research/nist_seed_p224_j_cube_20261025/report.md

Date: 2026-10-25. Scope: archived observations on the fixed 21-curve comparison (ten published controls, eleven reconstructed omissions, six base-field streams). Historical selection status: unresolved. This record preserves a specific lead; it does not establish that the original program required a cube root of j.

R59 | Pre-1997 FreeLIP validation on the fixed NIST/ANSI curve cohort

[R59] research/nist_seed_pre1997_library_20261027/report.md

2026-10-27. T-5518. This is an observation report, not a formal hypothesis decision. The missing historical selector remains unresolved. Recovered FreeLIP 1.0 passes the tested validation operations on all 11 reconstructed omissions and all 10 publications, across all six streams. These are reconstructed omissions, not documented rejection events.

R60 | Evidence review: NIST curve seed provenance (DEC-20261016-82753f)

[R60] research/nist_seed_provenance_20261016/review-DEC-20261016-82753f/report.md

Verified provenance, not cryptanalysis. This review changes no ECDLP cost, no attack claim, no curve-safety verdict and no SafeCurves cell. | | |---|---| | Decision | support, scoped (DEC-20261016-82753f) | | Evidence | EV-SCURVE-e9d71b: record-level replicated, per claim below | | Claim tier | crypto by parameter size (the standardized curves themselves). Scoped statements about those exact seeds only | | Review round | opened by DEC-20261016-7bcdd6; plan frozen in TASK-20261016-9bf74b at dbd1b8d15845579aaaa3af47b827955e2b625e77 | ...

ENTRIES 61-65 OF 70

Appendix A. Evidence catalog

Archival opening synopses below preserve each report's original scope. Earlier availability, planned-work and status statements may be superseded by the main narrative. These are internal research records, not authenticated original generator logs.

R61 | All-root SEA branch audit of the fixed NIST/ANSI cohort

[R61] research/nist_seed_sea_all_roots_20261027/report.md

2026-10-27. Descriptive execution and independent arithmetic observations. Historical selection remains unresolved.

R62 | Why did the eleven qualifying seed candidates go unpublished?

[R62] research/nist_seed_selection_deep_20261016/report.md

Date: 2026-10-16 Status: Historical selection cause unresolved; exact comparison and bounded twist-factorization diagnostics. Question: Do the eleven reported candidates passed over in the reconstructed prime-field seed streams fail a documented security rule, or exhibit a trait that explains why ten other curves were published?

R63 | P-224 seed-selection follow-up: the remaining twist case

[R63] research/nist_seed_selection_followup_20261016/report.md

Date: 2026-10-16 Scope: Four qualifying 20-byte blocks in the recovered P-224 stream; comparison with the previously checked 21 prime-field candidates. Status: Derived structural result; historical selection cause still unknown.

R64 | What might have selected the published curves?

[R64] research/nist_seed_selection_hypotheses_20261022/report.md

Date: 2026-10-22. Status: Exploratory, bounded comparison; historical selection remains unresolved. Question: What additional rule or process might explain the eleven mathematically qualifying seed blocks that occur in the recovered streams but were not published, and does the small generator space itself reveal a weak curve?

R65 | The ANSI base-point step and the eleven omitted prime-field curves

[R65] research/nist_seed_selection_workflow_20261023/report.md

Date: 2026-10-23 UTC. Status: Source-based and exact-record check; historical cause unresolved.

ENTRIES 66-70 OF 70

Appendix A. Evidence catalog

Archival opening synopses below preserve each report's original scope. Earlier availability, planned-work and status statements may be superseded by the main narrative. These are internal research records, not authenticated original generator logs.

R66 | Singular companions shared by the skipped and published curves

[R66] research/nist_seed_singular_companions_20261023/report.md

Date: 2026-10-23 UTC. Scope: Exact algebraic audit of the eleven reconstructed omitted seed classes and eight published controls over the same four prime fields. Result: This specific formula-compatible input surface is identical within each field and cannot separate the published and omitted labels.

R67 | The eleven skipped prime-field seed candidates

[R67] research/nist_seed_skipped_curves_20261016/report.md

Date: 2026-10-16 Question: What mathematical traits distinguish the qualifying curves that the recovered NIST/ANSI seed streams passed over from the curves that were published? Status: Reproducible structural assessment, supplemented by a check of a supplied 21-curve trait sheet. The historical reason for the omissions remains unknown.

R68 | Rational twist forms of the 21 NIST/ANSI seed curves

[R68] research/nist_seed_twist_classification_20261023/report.md

23 October 2026. Question: Under the working premise that the original program reached all eleven mathematically qualifying omitted seed curves, might they share a special quartic or sextic twist that the ten published curves lack?

R69 | Large quadratic-twist factors do not explain the eleven skipped curves

[R69] research/nist_seed_twist_embedding_20261023/report.md

Date: 2026-10-23 UTC. Status: bounded, retrospective arithmetic screen. Question: Supposing the original program evaluated all eleven omitted curves, do their quadratic twists have a large prime subgroup with unusually small pairing embedding degree?

R70 | NIST curve seed provenance: independent reproduction (2026-10-16)

[R70] research/nist_seed_provenance_20261016/README.md

Status. Verified provenance, not an attack. Nothing here changes the ECDLP on any curve. Source claim. Anthropic Frontier Red Team, Claude reveals ECDSA curve origins (anthropic.com/research/ecdsa-curve-origins), and F. Valsorda, NIST Elliptic Curves Seeds Bounty Has Been Claimed (words.filippo.io/seeds-solved), both 2026-10-16. The bounty was claimed on 2026-10-15 at 01:07:51 UTC. Claim. Every X9.62 / FIPS 186-2 prime-curve seed (P-192, P-224, P-256, P-384, P-521, prime192v2/v3, prime239v1-v3) is one 20-byte block of (int)(drand48() 256) output after srand48(t), with ...

ROWS 1-3 OF 21

Appendix B. Exact curve records

Canonical a=-3 short-Weierstrass model. All numerical field/model/order values below are hexadecimal. c is the SHA-derived parameter; t=p+1-n is the trace. Values are from the frozen cohort, not recomputed point counts.

239 bits | block 36 | skip-863711343-36 | omitted

stream: 863711343

seed: 25e56088185f8c3207b166cd2e007b88276afe78

UID: urn:ec-record:1:sha256:6eefbec5970878494b27923330575c21ea98ae6f88c91c0030998c81400a15db

p: 7fffffffffffffffffffffff7fffffffff8000000000007fffffffff

b: ab450c8bde1031f5fd172a7f697412567f6a1917f769a10192bd6f3de2a

c: 2c5818c45a189b2df97866b489caf669af7b2d39ecf5958d42149feb9818

n: 7fffffffffffffffffffffff7ffffef79acfcfd38003f45ebed6020532f3

239 bits | block 1562 | skip-863711343-1562 | omitted

stream: 863711343

seed: d09bfb7db85672dc700269f9189983f25f81958d

UID: urn:ec-record:1:sha256:360805ec2436c40bb27f999a3e28bfde9fac68cb1d7ee3388beb0872bdb56a88

p: 7fffffffffffffffffffffff7fffffffff8000000000007fffffffff

b: 639d458d9300d63240e743b4b1d4e7fa6bc78ef2af145b7b9139668db5b4

c: 7cc7e850007c1f87507fa16e0deabd4f28f169fc1cb7c1892b74c0459f7

n: 7fffffffffffffffffffffff8000002904acf44f113527eda41068f1981f

239 bits | block 2185 | prime239v1 | published

stream: 863711343

seed: e43bb460f0b80cc0c0b075798e948060f8321b7d

UID: urn:ec-record:1:sha256:05c3ea5b5307bc83a37a42e8a79bdeaed732b01b42d4fb274d3aaea8f6c536d5

p: 7fffffffffffffffffffffff7fffffffff8000000000007fffffffff

b: 6b016c3bdcf18941d0d654921475ca71a9db2fb27d1d37796185c2942c0a

c: 28b85ec1ecc19efe769eb741a6d1ba29476aa5a8f2610957d6efe78d3783

n: 7fffffffffffffffffffffff7fffff9e5e9a9f5d9071fbd1522688909d0b

ROWS 4-6 OF 21

Appendix B. Exact curve records

Canonical a=-3 short-Weierstrass model. All numerical field/model/order values below are hexadecimal. c is the SHA-derived parameter; t=p+1-n is the trace. Values are from the frozen cohort, not recomputed point counts.

239 bits | block 2320 | prime239v2 | published

stream: 863711343

seed: e8b4011604095303ca3b8099982be09fcb9ae616

UID: urn:ec-record:1:sha256:fa37ac33825f29e49eace88076d505bb9c04b946332fd29003c9e
bac7cddeceb

p: 7fffffffffffffffffffffff7fffffffffff8000000000007fffffffff

b: 617fab6832576cbbfed50d99f0249c3fee58b94ba0038c7ae84c8c832f2c

c: 1df491e44e7ccaf4d1ead8a6b90dae09e0d33f2c6cfe7a6ba76e86713d52

n: 7fffffffffffffffffffffff800000cfa7e8594377d414c03821bc582063

239 bits | block 2491 | skip-863711343-2491 | omitted

stream: 863711343

seed: 1256fdd46eeb96b7d312ddb1de99bff159cf6701

UID: urn:ec-record:1:sha256:7f20c02674790bb9cca864973434a3cd44eef6b2f57e8d22a8feb
88dd21db4aa

p: 7fffffffffffffffffffffff7fffffffffff8000000000007fffffffff

b: 999253d12139c2f7bbbfe2d836a41ceae53cf4d5e70f4b0bcf354acd2a4

c: 14a452622102b6907bb6e79a763199abdd21547597e8f2d84fda305bc755

n: 7fffffffffffffffffffffff80000036801f48c320794b8cb53fec0e008d

239 bits | block 2682 | skip-863711343-2682 | omitted

stream: 863711343

seed: d2c18e22558203444a2b33ea97c178675b438484

UID: urn:ec-record:1:sha256:04de63440b610d030a9534dbf1c97443d7e5eb39e39a1d165c6fe
39ec320a5ac

p: 7fffffffffffffffffffffff7fffffffffff8000000000007fffffffff

b: 45af05af6aac4dee0450d4b7794044398cbe9fcca12c9bccfb5cba03ee3

c: 2b70aefb6d37fcbb980c86523f7a9c415cc55b1ff478483fc48f96c8fd5f

n: 7fffffffffffffffffffffff8000001656326b616e478944dd218ec7fb07

ROWS 7-9 OF 21

Appendix B. Exact curve records

Canonical a=-3 short-Weierstrass model. All numerical field/model/order values below are hexadecimal. c is the SHA-derived parameter; t=p+1-n is the trace. Values are from the frozen cohort, not recomputed point counts.

239 bits | block 2863 | skip-863711343-2863 | omitted

stream: 863711343

seed: 3a700d3e6bf03255949804ba2a8677550c19ee11

UID: urn:ec-record:1:sha256:0e07eee7a888b4208fa6597b2d227d610eab07079b6df329a8fdb1ab1f7777ea

p: 7fffffffffffffffffffffff7fffffffff8000000000007fffffffff

b: 5e37d25ca9ef3d6a8ca8a95d08837a44b5c7de5c577ad2adbb5a4f6c782d

c: 2296d101b48b2f699fe7afbd77bf55be0d832184e9543f597b46fa0935d4

n: 7fffffffffffffffffffffff7fffff5e463b503ee96e521bf4194845d967

239 bits | block 3048 | prime239v3 | published

stream: 863711343

seed: 7d7374168ffe3471b60a857686a19475d3bfa2ff

UID: urn:ec-record:1:sha256:5422f32365cfedfd11fa09d91e55b32c202c11b78e44a4153e2c99b3e10c25b6

p: 7fffffffffffffffffffffff7fffffffff8000000000007fffffffff

b: 255705fa2a306654b1f4cb03d6a750a30c250102d4988717d9ba15ab6d3e

c: 3a4f9dc9a6cefd5f9d1193b9c9968c202430003c2819c2e498618dc58330

n: 7fffffffffffffffffffffff7fffff975deb41b3a6057c3c432146526551

192 bits | block 139 | prime192v1 | published

stream: 863966734

seed: 3045ae6fc8422f64ed579528d38120eae12196d5

UID: urn:ec-record:1:sha256:cdd1002da8b8992058e8b419334326a941a367e2e9298b81ba5759a58a3e1126

p: fffffffffffffffffffffffffffffffeffffffffffffffff

b: 64210519e59c80e70fa7e9ab72243049feb8deecc146b9b1

c: 3099d2bbbfcb2538542dcd5fb078b6ef5f3d6fe2c745de65

n: ffffffffffffffffffffffff99def836146bc9b1b4d22831

ROWS 10-12 OF 21

Appendix B. Exact curve records

Canonical a=-3 short-Weierstrass model. All numerical field/model/order values below are hexadecimal. c is the SHA-derived parameter; t=p+1-n is the trace. Values are from the frozen cohort, not recomputed point counts.

192 bits | block 2198 | skip-863966734-2198 | omitted

stream: 863966734

seed: 692d90186057951de011733105bf1b3d3b0fc451

UID: urn:ec-record:1:sha256:0bb52903786aa4f0ea0e77fdcd38fbd9e1f6580aa8afeb1d86e3d
f3dc6f7e33d

p: fffffffffffffffffffffffffffffffefffffffffffffff

b: beaa53dc228857db38331a9c08dbdce0e62baf417a2216e2

c: 4d60842d35e245e057ec2ae563d0734fc1a635d09687de41

n: ffffffffffffffffffffffffe5e830175c7011a77369b8549

192 bits | block 5228 | prime192v3 | published

stream: 863966734

seed: c469684435deb378c4b65ca9591e2a5763059a2e

UID: urn:ec-record:1:sha256:4bb29f37591675d48b0805ec04181b0c22237f21a56c9bf32078a
11b81238aa3

p: fffffffffffffffffffffffffffffffefffffffffffffff

b: 22123dc2395a05caa7423daeccc94760a7d462256bd56916

c: 25191f95024d839546d9a3375639a9967d52f1373bc4ee0b

n: ffffffffffffffffffffffff7a62d031c83f4294f640ec13

192 bits | block 5405 | skip-863966734-5405 | omitted

stream: 863966734

seed: 3ad10ff61797bc7258480cadd4e59c1ac1fb66d1

UID: urn:ec-record:1:sha256:db0cbd05fc040e764e213a561bf52691918385c835ad7fb52dfbe
5d5e5116192

p: fffffffffffffffffffffffffffffffefffffffffffffff

b: 912e4ef68a8df6948bb8e01d2d6dc8e0aff224187bbae807

c: 5510702babf5e0d141c6227e795825e246b8cbe4909085e5

n: ffffffffffffffffffffffff0e32ee481f43b57f75f5f9a1

ROWS 13-15 OF 21

Appendix B. Exact curve records

Canonical a=-3 short-Weierstrass model. All numerical field/model/order values below are hexadecimal. c is the SHA-derived parameter; t=p+1-n is the trace. Values are from the frozen cohort, not recomputed point counts.

192 bits | block 6013 | prime192v2 | published

stream: 863966734

seed: 31a92ee2029fd10d901b113e990710f0d21ac6b6

UID: urn:ec-record:1:sha256:6d3064894aabfbf742aeb34e6fe0f9d8b3cd487243e283adf7470ad08821f6a2

p: fffffffffffffffffffffffffffffffeffffffffffffff

b: cc22d6dfb95c6b25e49c0d6364a4e5980c393aa21668d953

c: 15038d1d2e1cafee0299f3011c1dc75b3c2a86e135db1e6b

n: ffffffffffffffffffffffe5fb1a724dc80418648d8dd31

521 bits | block 1136 | secp521r1 | published

stream: 871002724

seed: d09e8800291cb85396cc6717393284aaa0da64ba

UID: urn:ec-record:1:sha256:87a5992f487705d9729eb5c559a2e6667a448b8b9d8b71cf762a9607452bde36

p: 1ffffffffffffffffffffffffffffff

ROWS 16-18 OF 21

Appendix B. Exact curve records

Canonical a=-3 short-Weierstrass model. All numerical field/model/order values below are hexadecimal. c is the SHA-derived parameter; t=p+1-n is the trace. Values are from the frozen cohort, not recomputed point counts.

256 bits | block 1842 | prime256v1 | published

stream: 871696354

seed: c49d360886e704936a6678e1139d26b7819f7e90

UID: urn:ec-record:1:sha256:bb60dc2a4430532d22dc1a27ec2f7b78068bb317ede125594ce89219e65d1e4a

p: ffffffff00000001000000000000000000000000ffffffffffffffffffffffff

b: 5ac635d8aa3a93e7b3ebbd55769886bc651d06b0cc53b0f63bce3c3e27d2604b

c: 7efba1662985be9403cb055c75d4f7e0ce8d84a9c5114abcaf3177680104fa0d

n: ffffffff00000000ffffffffffffffffbce6faada7179e84f3b9cac2fc632551

384 bits | block 2688 | secp384r1 | published

stream: 872952608

seed: a335926aa319a27a1d00896a6773a4827acdac73

UID: urn:ec-record:1:sha256:d2e852530dd59cb7720c2e3af16270f51a4e294c89762848642da1331fdbc88f

p: fffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffeffffffff0000000000000000ffffffff

b: b3312fa7e23ee7e4988e056be3f82d19181d9c6efe8141120314088f5013875ac656398d8a2ed19d2a85c8edd3ec2aef

c: 79d1e655f868f02fff48dcdee14151ddb80643c1406d0ca10dfe6fc52009540a495e8042ea5f744f6e184667cc722483

n: ffffffffffffffffffffffffffffffffffffffffffffffffc7634d81f4372ddf581a0db248b0a77aecec196accc52973

224 bits | block 777 | skip-891607927-777 | omitted

stream: 891607927

seed: d00720e7691d403449c0d91be84bfb2c7501a267

UID: urn:ec-record:1:sha256:e3b59c25a3426411dad90ff57deac13f3f124d53d8bdf73622a03a0e8d406fe9

p: ffffffffffffffffffffffffffffffff000000000000000000000001

b: 5d66353d8430a14df0e4668aec2d6899234a1040bad2b4077ef9f344

c: 55af3812e2fa6dda812014f70925d9966c020ca9c65a16c9d182e189

n: ffffffffffffffffffffffffffff3fa279701925749a7de262aa8cc3

ROWS 19-21 OF 21

Appendix B. Exact curve records

Canonical a=-3 short-Weierstrass model. All numerical field/model/order values below are hexadecimal. c is the SHA-derived parameter; t=p+1-n is the trace. Values are from the frozen cohort, not recomputed point counts.

224 bits | block 1485 | skip-891607927-1485 | omitted

stream: 891607927

seed: 18cfdbcbb825568171212bc1610a5b8a8004e5c4

UID: urn:ec-record:1:sha256:b216c63c6aeb2cbb764ed35d5119701019998fe629a5ae2355206
2b570b3ae58

p: ffffffffffffffffffffffffffffffff000000000000000000000001

b: 579cd953cf8611029bd2ed436a48e02c4bfe614f9a66511374cf7a8d

c: e9071f373013662a51f8ed7defc1a83e656da6514a808530e47dd1

n: ffffffffffffffffffffffffffff819afa992d9c9222076db1ef949

224 bits | block 5736 | skip-891607927-5736 | omitted

stream: 891607927

seed: 3c170a30ee6709df9b4c9ea0b6c1e95fcd6200ce

UID: urn:ec-record:1:sha256:d9e812b141feb4a5dda20133dbbd21e59145d902c271418832e4a
7d48060a584

p: ffffffffffffffffffffffffffffffff000000000000000000000001

b: 4d41bdd4f0ada5e7746a76329e4fd5dd879e04adbfe7fcd2f5756748

c: 7d35027b9ca636a1113799dac7bd9a015ab4d0849f75ead0baca6825

n: ffffffffffffffffffffffffffff610362eb9e61800d49507468b249

224 bits | block 7026 | secp224r1 | published

stream: 891607927

seed: bd71344799d5c7fcdc45b59fa3b9ab8f6a948bc5

UID: urn:ec-record:1:sha256:db6c3a1e40ae241e8e4fb0695a6e68df7b430c74c4cc0a37e32da
95808e4df10

p: ffffffffffffffffffffffffffffffff000000000000000000000001

b: b4050a850c04b3abf54132565044b0b7d7bfd8ba270b39432355ffb4

c: 5b056c7e11dd68f40469ee7f3c7a7d74f7d121116506d031218291fb

n: ffffffffffffffffffffffffffff16a2e0b8f03e13dd29455c5c2a3d

SELECTED EXTERNAL DOCUMENTS

Appendix C. Primary-source reading map

Seed reconstruction disclosure - Six streams, published positions, eleven other qualifying windows; missing selection explanation.

https://words.filippo.io/seeds-solved/

Anthropic primary writeup - Reconstructed seed bytes and base-point recipes; open questions and limits on counting runs.

https://www.anthropic.com/research/ecdsa-curve-origins

17 November 1997 X9.62 draft - Retrieved full draft. Numerical generation procedure and seven published prime-field examples.

https://archive.org/download/x962-11-17-97/x962-11-17-97.pdf

20 September 1998-cover X9.62 text - Later generation and validation text; printed rounding differs from the 1997 draft.

https://safecurves.cr.yp.to/grouper.ieee.org/groups/1363/private/x9-62-09-20-98.pdf

Schoof 1995 - Point-counting framework and split/nonsplit information; not the original generator.

https://www.numdam.org/article/JTNB_1995__7_1_219_0.pdf

Morain 1995 - Practical point counting and the CCR coefficient-conversion correction.

https://www.numdam.org/article/JTNB_1995__7_1_255_0.pdf

Lehmann 1994 archived thesis - Modular-polynomial construction, interface and printed ell=5 witness.

https://web.archive.org/web/19970731071932id_/http://www.cs.uni-sb.de:80/users/mdenny/Bewohner/lehmann/lehmann.diplom.ps

SEC 1 version 2 - Later documentation of the MOV-bound increase from 20 to 100.

https://www.secg.org/sec1-v2.pdf

Baier 2002 thesis - Dated account of the CM class-number lower-bound advice.

https://d-nb.info/964515040/34

ETSI TS 102 176-1, July 2005 - Dated maximal-order class-number and operational ideal-class guidance.

https://www.etsi.org/deliver/etsi_ts/102100_102199/10217601/01.02.01_60/ts_10217601v010201p.pdf

RFC 5639, March 2010 - Later h(O_K)>10^7 requirement; no evidence of 1997 NIST adoption.

https://www.rfc-editor.org/rfc/rfc5639.html

Mattsson commit-pinned table - Modern 19-curve computation. Exact identity and SEA-convention reconciliation in R catalog.

https://github.com/emanjon/share/blob/6279fab0416e756998f0de6fcb0ccb473815127b/ecdsa_skipped_curves.txt

WHAT THIS PDF CONTAINS

Appendix D. Reproduction and custody

The source-manifest.json attachment records the source path and SHA-256 for every cataloged report. The source-report-bundle.zip attachment contains those internal Markdown reports plus the exact curve JSON, the final fixed-pass verification record, the Mattsson comparison JSON and the source manifest. The bundle does not redistribute the third-party thesis, standards PDFs, executable binaries or private source archives.

To inspect the Mattsson comparison, use research/nist_seed_mattsson_comparison_20261106/report.md and compare.py at commit 134317a08335f64746eef21d507fd2219f213c45. The report distinguishes the literal external cutoff, the integer output reconstruction, and the retained strict cutoff. The full historical investigation remains distinct from this comparison's validation.

To audit the closing pass, use research/nist_seed_final_checks_20261028/VERIFICATION.json, its report, FROZEN.json, INPUTS.json, and the retained runs/final-01 records. The verifier reports 63 certificates, 63 witnesses, 21 rows, one fresh cohort invocation and zero retries. Other report-specific archives and controls are located through the catalog links.

This synthesis is a document-production operation. It reads existing results, checks the stated population counts and constructs tables; it starts no point-counting, factorization, class-group, predicate-search, native historical-code or ECPP cohort trial. Internal source dates may be UTC while this cover uses the client date in America/Los_Angeles.

The report is not a claim that all untested arithmetic rules are impossible. It is not a claim that every omitted window was historically visited. It does not infer a weakness from a missing record. The evidence needed to settle the central question remains an authenticated acceptance mechanism or equivalent direct historical record explaining the actual choices.

Record identifiers

Canonical cohort SHA-256: e8d6a874e3dc77970c88957f4bca075a0cc7a431dd8bcb25e5c5d7bcba6eeb0f

Historical investigation source snapshot: d0b5c6e101400bde71f696e6c1cd215c0d9302e7

Mattsson publication commit: 134317a08335f64746eef21d507fd2219f213c45

Final assessment: historical selection criterion UNRESOLVED.

16 PAIRS / 19 TABLE INSTANCES / NO NEW SEARCH

Appendix E. Complete retained-fit catalog

This appendix reproduces every fitting pair in the archived ALL-FITS.json, in its original order. It validates only the nineteen recorded table applications on their existing 21-row signatures. It performs no feature generation, threshold tuning, pair enumeration or predicate search.

Counts: 16 signature pairs, 19 pair-specific truth tables, two distinct Boolean functions, 21 normalized signatures, and 54 named predicate aliases. There are 69 alias references across the pair slots because a signature is reused in several pairs. These are counts of representations, not counts of independent historical explanations.

Normalization and truth-table encoding

A raw predicate is a named feature followed by >= threshold or = value. An alias marked NOT(raw) is complemented to obtain the normalized signature bit. All truth tables below apply to normalized A and B, not automatically to the raw predicates. Aliases are equivalent only on these 21 observations (possibly after complementation); no universal mathematical equivalence is asserted.

For inputs ordered 00,01,10,11, output 1 means published and output 0 means omitted. The integer mask encodes output bit 2*A+B. Only masks 1 and 9 occur in the retained fits.

Mask / function	00	01	10	11	Applications
1 / NOR	1	0	0	0	13
9 / XNOR	1	0	0	1	6

F03, F04 and F16 have no cohort row in cell 11. NOR and XNOR therefore classify this cohort identically for those pairs while disagreeing on an unobserved input. This accounts for 19 table instances across 16 pairs. No claim of out-of-sample robustness follows from the archived use of the word robust.

Feature meanings and historical support

full_order is the full Frobenius matrix order; x_order quotients by global sign; projective_order is the order in PGL. Factor-degree and factor-count aliases refer to the derived division-polynomial profile. Known/unknown cost and rank features distinguish whether the splitting degree is supplied. A budget through L sums the recorded cost metric at tested primes <= L. character is the retained Kronecker/quadratic character, not the raw value modulo the prime. Rank refers to the correct candidate position, not a measured original runtime.

No pair below has verified historical use. The features can be connected to point-counting mathematics, but no recovered NIST/ANSI caller, threshold policy, run log or held-out historical decision authenticates these fitted selection rules. Twelve pairs explicitly name level 199; F16 uses a cumulative budget through 293, which also includes 199.

The exact retained fit JSON, selected signature masks and their extraction provenance are embedded and included in the reproducible build inputs. The source archive digest is e8c128fc7f78497775b83f2e8e49cbe6b92da9a12b7c6993ef293425619bd388.

ALL RETAINED FITS, UNRANKED

Appendix E. Pair and table index

Pair	Signature IDs A/B	Table IDs:mask	Aliases A/B	Historical use
F01	476 / 2291	T01:9	1 / 5	Unverified
F02	882 / 2003	T02:9	1 / 2	Unverified
F03	883 / 1999	T03:1, T04:9	2 / 5	Unverified
F04	884 / 1997	T05:1, T06:9	1 / 9	Unverified
F05	884 / 1998	T07:1	1 / 1	Unverified
F06	884 / 1999	T08:1	1 / 5	Unverified
F07	884 / 2000	T09:1	1 / 2	Unverified
F08	884 / 2001	T10:1	1 / 1	Unverified
F09	884 / 2002	T11:1	1 / 2	Unverified
F10	884 / 2004	T12:1	1 / 3	Unverified
F11	884 / 2046	T13:1	1 / 2	Unverified
F12	884 / 2048	T14:1	1 / 3	Unverified
F13	884 / 2245	T15:1	1 / 5	Unverified
F14	884 / 2271	T16:1	1 / 2	Unverified
F15	1069 / 2231	T17:9	1 / 1	Unverified
F16	1204 / 1859	T18:1, T19:9	2 / 3	Unverified

Pair numbers preserve source order; they are not a ranking by plausibility. Every complete table and every alias is printed on the following sixteen pages. A repeated signature ID refers to the same normalized 21-row bit vector, not a new independent measurement.

The representative cap rule C199<=32 and D199<=2048 in Chapter 11 is an interpretable description of a retained cohort split. Threshold aliases and other features can realize the same observed split. The truth tables below preserve the archived signature orientation rather than silently substituting a representative formula.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F01 - retained fitting pair

Normalized signatures: A=476, B=2291. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: NOT (signed/43/projective_order >= 14)

Representative B: signed/107/full_order >= 1272

A B	Published	Omitted	T01 / mask 9
00	7	0	1
01	0	8	0
10	0	3	0
11	3	0	1

T01: A equals B (XNOR); outputs [00,01,10,11] = 1,0,0,1.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/43/projective_order	>= 14	NOT(raw)
B	signed/107/full_order	>= 1272	raw
B	signed/107/x_order	>= 636	raw
B	signed/107/min_factor_degree	>= 636	raw
B	signed/107/max_factor_degree	>= 636	raw
B	signed/107/factor_count	>= 27	NOT(raw)

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. This is another in-sample fit, not an independently supported selection mechanism.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F02 - retained fitting pair

Normalized signatures: A=882, B=2003. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: signed/109/unknown_rank_max >= 16

Representative B: signed/227/known_rank_min >= 55

A B	Published	Omitted	T02 / mask 9
00	9	0	1
01	0	3	0
10	0	8	0
11	1	0	1

T02: A equals B (XNOR); outputs [00,01,10,11] = 1,0,0,1.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/109/unknown_rank_max	>= 16	raw
B	signed/227/known_rank_min	>= 55	raw
B	signed/227/unknown_rank_min	>= 55	raw

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. This is another in-sample fit, not an independently supported selection mechanism. Some aliases use the position of the correct eigenvalue candidate; that retrospective rank is not an observed historical timeout.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F03 - retained fitting pair

Normalized signatures: A=883, B=1999. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: signed/199/known_candidates_min >= 60

Representative B: signed/199/full_order >= 4400

A B	Published	Omitted	T03 / mask 1	T04 / mask 9
00	10	0	1	1
01	0	4	0	0
10	0	7	0	0
11 (unobserved)	0	0	0	1

T03: NOT A AND NOT B; outputs [00,01,10,11] = 1,0,0,0.

T04: A equals B (XNOR); outputs [00,01,10,11] = 1,0,0,1.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/199/known_candidates_min	>= 60	raw
A	signed/199/known_candidates_max	>= 60	raw
B	signed/199/full_order	>= 4400	raw
B	signed/199/x_order	>= 2200	raw
B	signed/199/min_factor_degree	>= 2200	raw
B	signed/199/max_factor_degree	>= 2200	raw
B	signed/199/factor_count	>= 10	NOT(raw)

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. Same-prime-199 relationship: linked to the retained lead, but the documented resource-policy follow-ups did not authenticate a historical gate.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F04 - retained fitting pair

Normalized signatures: A=884, B=1997. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: signed/199/projective_order >= 99

Representative B: signed/43/unknown_candidates_min >= 21

A B	Published	Omitted	T05 / mask 1	T06 / mask 9
00	10	0	1	1
01	0	2	0	0
10	0	9	0	0
11 (unobserved)	0	0	0	1

T05: NOT A AND NOT B; outputs [00,01,10,11] = 1,0,0,0.

T06: A equals B (XNOR); outputs [00,01,10,11] = 1,0,0,1.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/199/projective_order	>= 99	raw
B	signed/43/unknown_candidates_min	>= 21	raw
B	signed/43/unknown_candidates_max	>= 21	raw
B	signed/199/full_order	>= 4950	raw
B	signed/199/x_order	>= 2475	raw
B	signed/199/min_factor_degree	>= 2475	raw
B	signed/199/max_factor_degree	>= 2475	raw
B	signed/199/factor_count	>= 9	NOT(raw)
B	signed/269/known_rank_max	>= 63	raw
B	signed/269/unknown_rank_max	>= 64	raw

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. Includes an explicit level-199 feature with another retained feature; no original caller or fixed policy is identified. Some aliases use the position of the correct eigenvalue candidate; that retrospective rank is not an observed historical timeout.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F05 - retained fitting pair

Normalized signatures: A=884, B=1998. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: signed/199/projective_order >= 99

Representative B: signed/241/known_rank_max >= 18

A B	Published	Omitted	T07 / mask 1
00	10	0	1
01	0	2	0
10	0	8	0
11	0	1	0

T07: NOT A AND NOT B; outputs [00,01,10,11] = 1,0,0,0.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/199/projective_order	>= 99	raw
B	signed/241/known_rank_max	>= 18	raw

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. Includes an explicit level-199 feature with another retained feature; no original caller or fixed policy is identified. Some aliases use the position of the correct eigenvalue candidate; that retrospective rank is not an observed historical timeout.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F06 - retained fitting pair

Normalized signatures: A=884, B=1999. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: signed/199/projective_order >= 99

Representative B: signed/199/full_order >= 4400

A B	Published	Omitted	T08 / mask 1
00	10	0	1
01	0	2	0
10	0	7	0
11	0	2	0

T08: NOT A AND NOT B; outputs [00,01,10,11] = 1,0,0,0.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/199/projective_order	>= 99	raw
B	signed/199/full_order	>= 4400	raw
B	signed/199/x_order	>= 2200	raw
B	signed/199/min_factor_degree	>= 2200	raw
B	signed/199/max_factor_degree	>= 2200	raw
B	signed/199/factor_count	>= 10	NOT(raw)

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. Same-prime-199 relationship: linked to the retained lead, but the documented resource-policy follow-ups did not authenticate a historical gate.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F07 - retained fitting pair

Normalized signatures: A=884, B=2000. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: signed/199/projective_order >= 99

Representative B: signed/227/known_rank_max >= 105

A B	Published	Omitted	T09 / mask 1
00	10	0	1
01	0	2	0
10	0	8	0
11	0	1	0

T09: NOT A AND NOT B; outputs [00,01,10,11] = 1,0,0,0.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/199/projective_order	>= 99	raw
B	signed/227/known_rank_max	>= 105	raw
B	signed/227/unknown_rank_max	>= 105	raw

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. Includes an explicit level-199 feature with another retained feature; no original caller or fixed policy is identified. Some aliases use the position of the correct eigenvalue candidate; that retrospective rank is not an observed historical timeout.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F08 - retained fitting pair

Normalized signatures: A=884, B=2001. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: signed/199/projective_order >= 99

Representative B: signed/241/known_rank_min >= 7

A B	Published	Omitted	T10 / mask 1
00	10	0	1
01	0	2	0
10	0	8	0
11	0	1	0

T10: NOT A AND NOT B; outputs [00,01,10,11] = 1,0,0,0.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/199/projective_order	>= 99	raw
B	signed/241/known_rank_min	>= 7	raw

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. Includes an explicit level-199 feature with another retained feature; no original caller or fixed policy is identified. Some aliases use the position of the correct eigenvalue candidate; that retrospective rank is not an observed historical timeout.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F09 - retained fitting pair

Normalized signatures: A=884, B=2002. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: signed/199/projective_order >= 99

Representative B: signed/241/known_rank_min >= 4

A B	Published	Omitted	T11 / mask 1
00	10	0	1
01	0	2	0
10	0	7	0
11	0	2	0

T11: NOT A AND NOT B; outputs [00,01,10,11] = 1,0,0,0.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/199/projective_order	>= 99	raw
B	signed/241/known_rank_min	>= 4	raw
B	signed/241/known_rank_max	>= 12	raw

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. Includes an explicit level-199 feature with another retained feature; no original caller or fixed policy is identified. Some aliases use the position of the correct eigenvalue candidate; that retrospective rank is not an observed historical timeout.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F10 - retained fitting pair

Normalized signatures: A=884, B=2004. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: signed/199/projective_order >= 99

Representative B: signed/241/known_candidates_min >= 32

A B	Published	Omitted	T12 / mask 1
00	10	0	1
01	0	2	0
10	0	6	0
11	0	3	0

T12: NOT A AND NOT B; outputs [00,01,10,11] = 1,0,0,0.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/199/projective_order	>= 99	raw
B	signed/241/known_candidates_min	>= 32	raw
B	signed/241/known_candidates_max	>= 32	raw
B	signed/241/known_rank_max	>= 7	raw

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. Includes an explicit level-199 feature with another retained feature; no original caller or fixed policy is identified. Some aliases use the position of the correct eigenvalue candidate; that retrospective rank is not an observed historical timeout.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F11 - retained fitting pair

Normalized signatures: A=884, B=2046. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: signed/199/projective_order >= 99

Representative B: signed/budget/31/unknown_candidates_min >= 37

A B	Published	Omitted	T13 / mask 1
00	10	0	1
01	0	2	0
10	0	7	0
11	0	2	0

T13: NOT A AND NOT B; outputs [00,01,10,11] = 1,0,0,0.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/199/projective_order	>= 99	raw
B	signed/budget/31/unknown_candidates_min	>= 37	raw
B	signed/budget/31/unknown_candidates_max	>= 37	raw

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. Includes an explicit level-199 feature with another retained feature; no original caller or fixed policy is identified.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F12 - retained fitting pair

Normalized signatures: A=884, B=2048. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: signed/199/projective_order >= 99

Representative B: signed/budget/31/unknown_candidates_min >= 35

A B	Published	Omitted	T14 / mask 1
00	10	0	1
01	0	2	0
10	0	6	0
11	0	3	0

T14: NOT A AND NOT B; outputs [00,01,10,11] = 1,0,0,0.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/199/projective_order	>= 99	raw
B	signed/budget/31/unknown_candidates_min	>= 35	raw
B	signed/budget/31/unknown_candidates_max	>= 35	raw
B	signed/budget/31/unknown_rank_max	>= 28	raw

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. Includes an explicit level-199 feature with another retained feature; no original caller or fixed policy is identified. Some aliases use the position of the correct eigenvalue candidate; that retrospective rank is not an observed historical timeout.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F13 - retained fitting pair

Normalized signatures: A=884, B=2245. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: signed/199/projective_order >= 99

Representative B: signed/283/full_order >= 80088

A B	Published	Omitted	T15 / mask 1
00	10	0	1
01	0	2	0
10	0	8	0
11	0	1	0

T15: NOT A AND NOT B; outputs [00,01,10,11] = 1,0,0,0.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/199/projective_order	>= 99	raw
B	signed/283/full_order	>= 80088	raw
B	signed/283/x_order	>= 40044	raw
B	signed/283/min_factor_degree	>= 40044	raw
B	signed/283/max_factor_degree	>= 40044	raw
B	signed/283/factor_count	>= 3	NOT(raw)

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. Includes an explicit level-199 feature with another retained feature; no original caller or fixed policy is identified.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F14 - retained fitting pair

Normalized signatures: A=884, B=2271. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: signed/199/projective_order >= 99

Representative B: signed/11/known_rank_max >= 4

A B	Published	Omitted	T16 / mask 1
00	10	0	1
01	0	2	0
10	0	7	0
11	0	2	0

T16: NOT A AND NOT B; outputs [00,01,10,11] = 1,0,0,0.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/199/projective_order	>= 99	raw
B	signed/11/known_rank_max	>= 4	raw
B	signed/11/unknown_rank_max	>= 4	raw

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. Includes an explicit level-199 feature with another retained feature; no original caller or fixed policy is identified. Some aliases use the position of the correct eigenvalue candidate; that retrospective rank is not an observed historical timeout.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F15 - retained fitting pair

Normalized signatures: A=1069, B=2231. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: NOT (signed/263/projective_order >= 132)

Representative B: NOT (character/twist_order/173 = 1)

A B	Published	Omitted	T17 / mask 9
00	4	0	1
01	0	6	0
10	0	5	0
11	6	0	1

T17: A equals B (XNOR); outputs [00,01,10,11] = 1,0,0,1.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/263/projective_order	>= 132	NOT(raw)
B	character/twist_order/173	= 1	NOT(raw)

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. This is another in-sample fit, not an independently supported selection mechanism.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.

COMPLETE ALIASES AND PAIR-SPECIFIC TRUTH TABLES

Appendix E. F16 - retained fitting pair

Normalized signatures: A=1204, B=1859. Every retained table assigns all 10 publications output 1 and all 11 omissions output 0 on the archived cohort.

Representative A: signed/budget/293/known_candidates_min >= 1001

Representative B: character/trace/7 = 0

A B	Published	Omitted	T18 / mask 1	T19 / mask 9
00	10	0	1	1
01	0	3	0	0
10	0	8	0	0
11 (unobserved)	0	0	0	1

T18: NOT A AND NOT B; outputs [00,01,10,11] = 1,0,0,0.

T19: A equals B (XNOR); outputs [00,01,10,11] = 1,0,0,1.

All aliases (not independent fits)

Bit	Named feature	Predicate	Normalized bit
A	signed/budget/293/known_candidates_min	>= 1001	raw
A	signed/budget/293/known_candidates_max	>= 1001	raw
B	character/trace/7	= 0	raw
B	ecpp/cm_steps	>= 6	NOT(raw)
B	signed/7/projective_order	>= 4	NOT(raw)

Historical support and interpretation

No retrieved source establishes historical use of this exact pair, threshold choices, or truth table. This is another in-sample fit, not an independently supported selection mechanism. The cumulative budget through 293 includes level 199; it is not an independent test excluding that level.

Evidence status: exact retained cohort fit; aliases are observational; historical criterion UNRESOLVED. No additional search was performed for this appendix.